

On the Determinants of Cooperation in Infinitely Repeated Games: A Survey

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Abstract

There have been major advances in the theoretical study of the determinants of cooperation with many applications in economics and the social sciences. A growing experimental literature has studied the determinants of cooperation, tested different predictions of the theory, and suggested an empirical solution to the problem of multiple equilibria in infinitely repeated games. To provide a robust description of the literature's findings, we gather and analyze a meta-data set of experiments on infinitely repeated prisoners' dilemma games. The experimental data shows that cooperation is affected by infinite repetition and that cooperation is more likely to arise when it can be supported in equilibrium. However, the fact that cooperation can be supported in equilibrium does not imply that most subjects will cooperate. High cooperation rates will only emerge when the parameters of the repeated game are such that cooperation is very robust to strategic uncertainty. We also review the results regarding the effect of imperfect monitoring, changing partners and personal characteristics on cooperation and the strategies used to support it. *JEL codes:* C7, C9.

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1 Introduction

The tension between opportunistic behavior and cooperation is a central feature of human interactions. Understanding whether and how people can overcome the incentives for opportunistic behavior and cooperate is important to economics and other sciences. One of the contributions of game theory to the study of this tension is to recognize the fact that repeated interaction enable credible punishments and rewards that can lead to cooperation.

There have been major advances in the theory of infinitely repeated games, studying the determinants of cooperation in increasingly complex environments, from perfect monitoring to imperfect monitoring (both public and private) and others. Applications are many, both in economics and other social sciences. However, the theory of infinitely repeated games is sometimes criticized for not providing sharp predictions (when players are sufficiently patient both cooperation and defection are possible equilibrium actions).¹ Hence, empirical investigations are important, not only to test theoretical predictions, but also to help us sharpen predictions when theory provides multiple ones. Empirical studies of repeated games have been slower to emerge as many of the important parameters are difficult to observe, such as the discount factor, the probability that an interaction will continue, or the information that agents have about the choices of others. This makes experimental investigation a particularly useful method to test, explore, and refine these models. Systematically studying how agents behave in such environments help us learn about the determinants of cooperation.

After a period of initial investigation from the 1970s to 2000 that included only a handful of studies, the last decade has seen a significant increase in the number of laboratory investigations into infinitely repeated interactions: games where subjects interact for a number of periods without a known termination period and the experimenter controls the discount factor. This is most typically induced using a random termination period, where all subjects know

¹Tirole [1988] says that “[t]he multiplicity of equilibria is an embarrassment of riches” and Fudenberg and Maskin [1993] state that “[t]he theory of repeated games has been somewhat disappointing. [...] the theory does not make sharp predictions.”

the probability with which the game will continue for an additional period; a method first used by Roth and Murnighan [1978].²

We review the experimental literature on the determinants of cooperation with a special focus on the prisoner's dilemma. The prisoner's dilemma captures the tension at the heart of social dilemmas in its simplest form; and as such it is a useful tool to explore general questions about cooperation. Furthermore, we will go beyond the simple review and use meta-data assembled from previous experimental studies to revisit the key findings of this literature. For this purpose we need a common game and the prisoner's dilemma is the one that has been most studied with random termination in the laboratory.

The use of meta-data to revisit the main findings of the literature is useful as the design of each experiment is usually specifically tailored to highlight a certain comparative static or element. The knowledge we accumulate this way is often precise, but on the other hand we rarely have a sense of its robustness. How dependent, for instance, is a specific result to the constellation of parameters used in an experiment? To the subject pool? To the details of the procedures and the formulation of the instructions? The hope is that such questions are resolved over time through replication with different payoffs, instructions, subjects, and procedures used by other researchers. However it is often difficult to get a good sense of the robust results that come out of a sequence of experiments. It is with this in mind that we use this meta-data to re-examine some of the questions that others, and ourselves, have studied in specific settings.³

We start, in section 2, by considering environments with perfect monitoring and discuss how the parameters of the game affect cooperation, the evolution

²Some authors prefer to talk about indefinitely, as opposed to infinitely, repeated games given the method used to implement them. We view the term infinitely repeated games as a reference to the theoretical framework used to analyze the situation of interest and not so much as a description of the implementation. Accordingly, we will use the terminologies randomly terminated games, indefinite games, and infinitely repeated games interchangeably.

³Meta-studies typically gather the results of multiple articles. Although we will do that in some subsections, much of the meta-analysis is done by re-analyzing the raw data from multiple studies that we obtained and pooled together (as well as standardized).

of cooperation as subjects gain experience and the choice of strategies. This is followed by a short section on the methods used to induced infinitely repeated games in the laboratory and potential implementation issues. In section 4 we discuss the impact of personal characteristics on cooperation and the motives behind cooperation. We then study cooperation under imperfect monitoring in section 5. In section 6 we present evidence on community enforcement when partners may change from round to round. In section 7 we describe work using stage games other than the prisoners' dilemma and more general dynamic games. The last section explores some promising directions for future research.

2 The Infinitely Repeated Prisoners' Dilemma Game With Perfect Monitoring

In this section we address the main questions raised by the literature by focusing on the simplest type of infinitely repeated game: infinitely repeated prisoners' dilemma (henceforth PD) games under perfect monitoring and fixed matching. Treatments in these papers mainly differ in the probability of continuation (δ) and the payoff matrix.⁴

A PD payoff matrix consists of four numbers, the temptation payoff (T) which is the payoff to defecting when the other cooperates, the reward from joint cooperation (R), the punishment from mutual defection (P), and the sucker's payoff from cooperation when the other defects (S). For these payoffs to define a PD, it must be that $T > R > P > S$. Furthermore, it is often required that $2R > T + S$ so that alternating between cooperation and defection cannot be more profitable than joint cooperation.

To facilitate comparison across treatments we will work with the normalized matrix in Table 1 which reduces the number of payoff parameters to two: g is the gain from defection when the other player cooperates and ℓ is the loss from cooperation when the other player defects. This matrix is obtained

⁴Other differences have to do with factors outside of the standard theory such as the number of supergames in a session, instructions, and feedback between supergames.

Table 1: Prisoners' Dilemma Row Player's Payoffs

Original			Normalized		
	C	D		C	D
C	R	S	C	$\frac{R-P}{R-P} = 1$	$\frac{S-P}{R-P} = -\ell$
D	T	P	D	$\frac{T-P}{R-P} = 1 + g$	$\frac{P-P}{R-P} = 0$

Table 2: Percentage of Cooperation (First Rounds)

	Probability of Continuation		
	0.105	0.5	0.895
Roth and Murnighan (1978)	19	29.75	36.36
Murnighan and Roth (1983)	17.83	37.48	29.07

by applying a monotonic linear transformation to the original matrix which, under general utility assumptions, should not affect behavior.⁵

2.1 Does Repetition Matter?

Roth and Murnighan [1978] and Murnighan and Roth [1983] were the first to address experimentally the issue of whether infinite repetition affects cooperation. They introduced the random termination rule and had subjects play against an artificial opponent (a pre-programmed strategy) for one repeated game per treatment for money.

As can be seen in Table 2, cooperation tends to increase with the probability of continuation but the effect is not always monotonic in those papers. Moreover, even under high probability of continuation, the subjects are far away from making the most of the opportunity to cooperate. This lead Roth [1995] to conclude that “the results remain equivocal.”

Feinberg and Husted [1993] combined a random continuation rule (following a fixed number of rounds played for certain) with discounting of payoffs (in every period) in a repeated prisoners' dilemma game. They find that less discounting resulted in an increase of cooperative outcomes. However, the effect of repetition was not large.

⁵We discuss the empirical validity of this assumption in footnote 8.

Finally, Palfrey and Rosenthal [1994] studied contributions to a public good under incomplete information regarding the cost of contribution. They find that contribution rates are greater under a probability 0.9 of continuation than in one-shot games. However, the magnitude of the increase was not large, leading them to conclude that the impact of repetition on cooperation is limited.

A second wave of experimental papers, starting with Dal Bó [2005], provides a more positive answer on the issue of whether *the shadow of the future* can increase cooperation and by how much. Dal Bó [2005] finds that when going from one shot PD games to infinitely repeated ones with $\delta = .75$ cooperation increases fourfold.

We will use the data generated by this second wave of experiments to describe the evidence on the effect of repetition on cooperation and the other main results in the literature. In this way, the results we present will be robust in the sense that they are not based on one particular paper but on the data from several experiments conducted by different experimenters in different places with different procedures, instructions, and subject pools.

Table 3 displays the variety of treatments used in the second wave of experiments for which we have data (papers are organized by date of publication, then treatments are ordered by their parameters δ , g and l). In this section we focus on randomly terminated PD experiments with perfect monitoring from nine papers. None of the treatments included combine random termination with other forms of discounting (nor do they vary discounting within a supergame). We also include two articles with one-shot PD games: $\delta = 0$.

The information in Table 3 is only for the treatments we use in this section; the papers in which these were originally used might have had many more sessions or treatments, but those did not fit our requirements (deterministic PD with perfect monitoring, fixed and known continuation probability, played in a fixed pair). Also, some of the studies used within-subjects designs; in those cases we only include the parameters before the first change in treatment. This is to avoid any “contamination” across treatments that we would not account for.

We have data from eleven papers, involving 28 different treatments (com-

Table 3: General Information About Data Sets Used

	Sessions	Subjects	δ	g	ℓ	Supergames
Andreoni and Miller 1993	1	14	0	1.67	1.33	200
Dal Bó 2005	6	276				
	2	72	0	1.17	0.83	7
	2	102	0	0.83	1.17	9
	1	42	0.75	1.17	0.83	7
	1	60	0.75	0.83	1.17	10
Aoyagi and Fréchette 2009	4	74				
	2	36	0	0.33	0.11	75
	2	38	0.9	0.33	0.11	10
Duffy and Ochs 2009	9	102	0.9	1	1	13
Dal Bó, Foster, Putterman 2010	28	424	0	1	3	10
Dal Bó and Fréchette 2011	18	266				
	3	44	0.5	2.57	1.86	71
	3	50	0.5	0.67	0.87	72
	3	46	0.5	0.09	0.57	77
	3	44	0.75	2.57	1.86	33
	3	38	0.75	0.67	0.87	47
	3	44	0.75	0.09	0.57	35
Blonski, Ockenfels, and Spagnolo 2011	10	200				
	1	20	0.5	2	2	11
	2	40	0.75	2	2	11
	1	20	0.75	1	8	11
	1	20	0.75	1	1	11
	1	20	0.75	0.83	0.5	11
	1	20	0.75	0.75	1.25	11
	1	20	0.75	0.5	3.5	11
	1	20	0.875	2	2	8
	1	20	0.875	0.5	3.5	8
Fudenberg, Rand, Dreber 2012	3	48	0.875	0.33	0.33	9
Sherstyuk, Tari, Saijo 2013	4	56	0.75	1	0.25	29
Dal Bó and Fréchette 2013	35	554				
	3	50	0.5	2.57	1.86	37
	8	140	0.5	0.09	0.57	46
	8	114	0.75	2.57	1.86	25
	8	120	0.75	0.09	0.57	24
	8	130	0.9	2.57	1.86	12
Fréchette and Yuksel 2014	3	60	0.75	0.4	0.4	12
Total	121	2064		Choices: 122, 233		

binations of δ , g and ℓ), with 2064 subjects and a total of 122,233 choices. An important difference to the first wave of experiments is that the articles in Table 3 had subjects participating in several supergames under the same combination of parameters thus allowing them to gain experience. Note that the treatments with the smallest number of supergames have 7 supergames.

Remember that the first wave of experiments on infinitely repeated games involved playing a single supergame and only found a modest relation between the probability of continuation and cooperation. We can perform a similar exercise using data from the first supergame of each study in Table 3.

We focus on round 1 choices for most of the analysis.⁶ While behavior in round 1 provides a somewhat incomplete picture of choices in a supergame, it simplifies the analysis for two reasons. First, supergames will have different number of rounds due to the random termination rule making comparisons that use all rounds difficult. Second, behavior in higher rounds is not independent of previous rounds requiring careful consideration of this dependence. Later in the paper, choices in the supergame as a whole will be analyzed.

The first two columns in Table 4 present the results of a probit of cooperation in round 1 of the first supergame on normalized payoffs and discount factor.⁷ Cooperation in the first round of the first supergame is decreasing in both g and ℓ and increasing in δ . However, an increase in the probability of continuation δ only has a small effect, consistent with the evidence from the first wave of experiments.⁸

⁶The term round is used to denote a play of the stage game. It is also sometimes referred to as a period. A play of a supergame is often called match or cycle.

⁷Standard errors are clustered at the paper level to account for possible effects of instructions, procedures, or software. An additional source of correlation could be session-effects. Given that clustering by paper allows for arbitrary correlations within paper, it does as well within session. For a discussion of session-effects see Fréchette [2011]. In cases where there are multiple observations per subject we will allow for random-effects at the subject level. If, in addition, there are lagged regressors, we will then use a correlated random-effects specification to deal with the initial conditions problem (see Heckman [1981] or Chamberlain [1982] for the static case and Wooldridge [2002] for a clear exposition of the initial conditions problem and methods to address it).

⁸Remember that the normalization of payoffs presented in Table 1 depends on behavior not being affected by linear monotonic transformation of payoffs. To assess the validity

Table 4: The Effect of Repetition on Round 1 Cooperation

	Supergame 1		Supergame 7		Supergame 15	
	Probit	Marginal	Probit	Marginal	Probit	Marginal
	(1)	(2)	(3)	(4)	(5)	(6)
g	-0.206*** (0.0323)	-0.0811*** (0.0128)	-0.320*** (0.0772)	-0.114*** (0.0299)	-0.291* (0.167)	-0.113* (0.0643)
ℓ	-0.154*** (0.0386)	-0.0607*** (0.0154)	-0.235** (0.101)	-0.0842** (0.0352)	-0.490 (0.320)	-0.191 (0.125)
δ	0.378** (0.147)	0.149*** (0.0575)	1.282*** (0.304)	0.458*** (0.104)	2.064** (0.848)	0.804** (0.325)
Constant	0.126 (0.132)		-0.389 (0.239)		-0.650 (0.516)	
N	2064		1970		748	

Clustered Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The second two columns in Table 4 present the results of a probit of cooperation in round 1 of the seventh supergame on normalized payoffs and discount factor. We choose the seventh supergame as this is the highest supergame we can study without losing any treatment (combination of δ , g and ℓ). The last two columns in Table 4 present the results of a probit of cooperation in round 1 of supergame 15 on normalized payoffs and discount factor. We show results from this supergame to confirm that the results on the effect of experience on behavior are not limited to supergame 7. The last four columns in Table 4 show that, after subjects have gained experience playing supergames, the effect of the probability of continuation is much larger than in the first supergame.

Using the results in Table 4, we can compute the predicted probability of cooperation for the parameters used by Roth and Murnighan [1978].⁹ Table 5 repeats the main result of Roth and Murnighan [1978] and includes the predicted probabilities from the estimation performed on the data from the new wave of experiments. While the predicted levels based on the new estimates are far from the original cooperation rates in round 1 of the first supergame, the qualitative effect of δ on cooperation is very similar: increasing the continuation probability only leads to a small increase in cooperation.¹⁰ The original experiment reports that increasing δ from 0.105 to 0.895 leads to about a 17 percentage point increase in cooperation. The new estimates predict about a 12 percentage point increase in the first supergame. Hence, in both cases,

of this assumption we use a similar analysis to the one presented in Table 4 in which we include as controls the multiplicative and additive factors used for the transformation in each treatment. We find that these factors are not statistically significant in supergame 1 and 15 and the multiplicative factor is only significant at the 10 percent level in supergame 7. This suggest that the representation of the PD games with only two parameters, g and ℓ , is appropriate. Consistently, Kagel and Schley [2013] find that cooperation is not affected by expressing payoffs in cents instead of dollars.

⁹The normalized game has $g = \ell = 0.5$.

¹⁰The differences in levels between the observed behavior and the predicted one may be the result of two important differences in the way the original experiment was conducted relative to those in Table 3. First, subjects played against computerized opponents. Second, subjects were not paid proportionally to their performance. Rather, a prize was given to the “best player.” Given those differences, it might not be so surprising that levels are different.

there is an impact of rather limited magnitude.

Table 5: Cooperation in Old and New Experiments (Round 1)

	Continuation Probability		
	0.105	0.5	0.895
Roth and Murnighan	19.00	29.75	36.36
New Estimates (fitted):			
Supergame 1	49.42	55.37	61.19
Supergame 7	29.73	48.97	68.46
Supergame 15	20.50	49.66	79.01

Given the analysis based on the new wave of experiment in Table 4, what would have Roth and Murnighan [1978] found if they had allowed subjects to gain experience? The predicted cooperation rates from the last two rows in Table 5 show that they would have found a much greater effect of repetition on cooperation. The predicted levels show that increasing δ from 0.105 to 0.895 leads to about a 39 percentage point increase in cooperation in supergame 7 and 59 percentage points in supergame 15. This suggests that Roth and Murnighan [1978] would have found a greater impact of repetition on cooperation if their subjects had interacted in several repeated games.

To further explore how experience affects the impact of repetition on cooperation, we estimate a regression such as the ones reported in Table 4 for each supergame. Figure 1 presents the predicted cooperation rates for supergames one to 15, again assuming the payoffs of Roth and Murnighan [1978], for a δ of 0 and 0.9; two extreme values for which we have data.¹¹ By supergame 15 the impact of δ is substantial. In fact, the marginal effect of δ when the payoffs are set at their average values in the sample is 0.8. Hence, having repeated interactions with an uncertain horizon has an important impact on the ability of subjects to support cooperation once they have gained experience.

¹¹After supergame 15 the sample size diminishes rapidly. Results past supergame seven should be interpreted with the caveat that the sample starts to change. We note, however, that the upward trend for $\delta = 0.9$ and the downward trend for $\delta = 0$ are present from the very start.

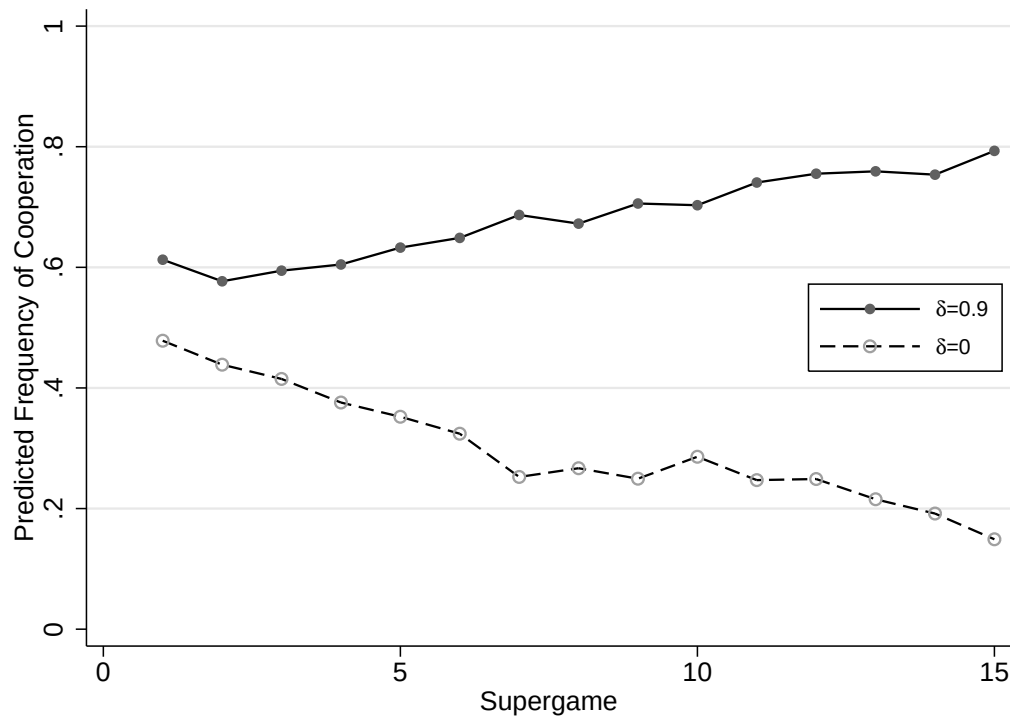


Figure 1: The Impact of δ on Round 1 Cooperation by Supergame

Two potential concerns with respect to the interpretation of Figure 1 are the focus on round one and on cooperation decisions as opposed to cooperative outcomes (joint cooperation). Indeed, in finitely repeated games, round one cooperation rates are often much higher than last round cooperation. Some may think that a similar phenomenon (an important decrease in cooperation rates over time) occurs in randomly terminated games. Once one considers rounds beyond the first, the rate of cooperative outcomes is not simply the product of cooperative choices since choices in a pair are no longer independent. Hence, cooperation rates that are relatively high could hypothetically hide much lower cooperative outcomes. Figure 2 presents the predicted rates of joint cooperation in the last round of a supergame; again for the same normalized payoffs ($g = \ell = 0.5$). It shows that the impact of δ on cooperation is not limited to the first round but persists until the end of the supergames.

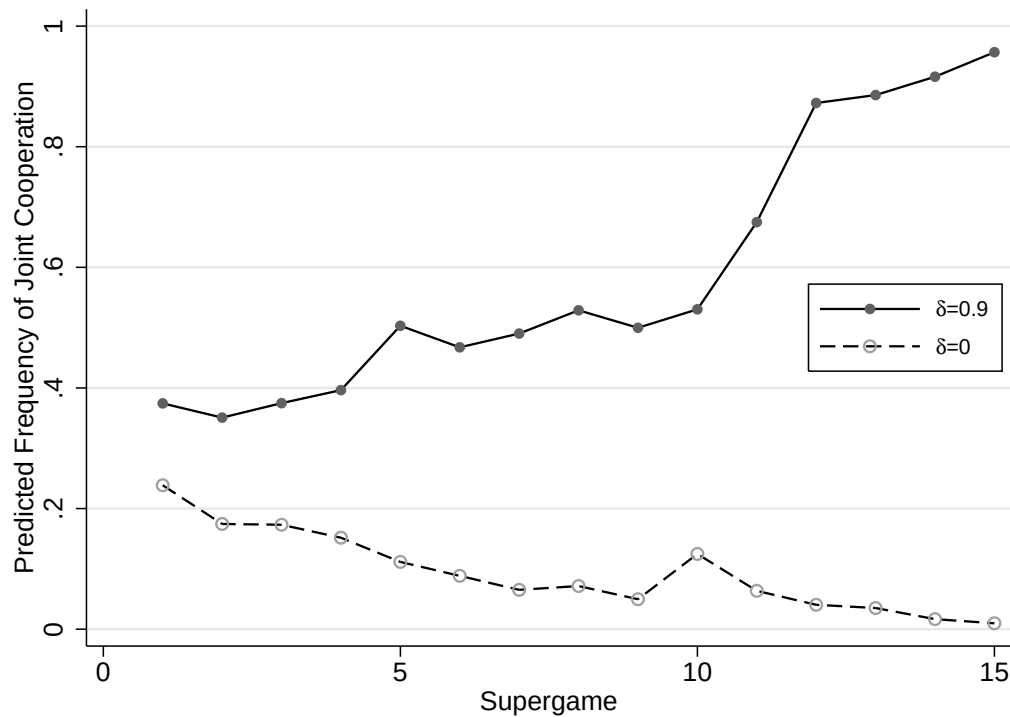


Figure 2: The Impact of δ on Last Round Joint Cooperation by Supergame

The analysis presented so far leads us to the following result.

Result 1 *Cooperation is increasing in the probability of future interactions and this effect increases with experience.*

2.2 The Predictive Power Of Theory

The fact that having randomly terminated repeated interactions affects behavior does not necessary mean that it does as theory predicts. Testing models of infinitely repeated games is complicated by the fact that many outcomes are consistent with theoretical predictions for large probability of continuations due to the “folk theorem:” any feasible and individually rational payoff can

be supported in equilibrium if players are sufficiently patient.¹²

However, the theory does have testable predictions. For example, cooperation rates should not be higher when cooperation cannot be supported in equilibrium than when it can. Several articles have studied how cooperation depends on whether mutual cooperation can be supported in an equilibrium under the assumption of monetary payoffs and risk neutrality.¹³

For any payoff matrix we can calculate the minimum δ required to support mutual cooperation in a subgame perfect equilibrium (SPE):

$$\delta^{SPE} = \frac{g}{1+g}. \quad (1)$$

The greater the temptation to defect (g) is, the greater the shadow of the future must be for subjects to have an incentive to cooperate.¹⁴

Roth and Murnighan [1978] and Murnighan and Roth [1983] were the first to show that cooperation levels are greater under treatments for which cooperation can be supported in equilibrium. This question is revisited in the new wave of articles. In particular, Dal Bó and Fréchette [2011] show that cooperation declines to negligible levels when δ is such that cooperation cannot be supported in equilibrium while it is higher when cooperation can be supported as part of a SPE.¹⁵ We use the data from the new wave of experiments to study

¹²On the “folk theorem” under perfect monitoring see Friedman [1971] and Fudenberg and Maskin [1986]. For a description of the equilibrium payoffs for infinitely repeated PD games for each discount factor see Stahl [1991]. For a review of the theoretical literature on infinitely repeated games see Mailath and Samuelson [2006].

¹³We exclude for now cases with equilibria in which alternating cooperation can be supported but not mutual cooperation.

¹⁴This critical value δ^{SPE} is calculated as follows. Assume that the other player is following the strategy Grim: cooperate in round 1 and continue cooperating until there is a defection, then defect forever. The strategy Grim provides the strongest possible punishment for defection. Under the assumption that the other player is following Grim, a player would obtain a payoff of $\frac{1}{1-\delta}$ by also choosing the strategy Grim (or any other cooperative strategy), while he would obtain a payoff of $1+g$ by defecting. Therefore, if the other player is following Grim, a player has an incentive to cooperate only if $\delta \geq \frac{g}{1+g}$.

¹⁵This question can also be evaluated in Aoyagi and Fréchette [2009] who report results for $\delta = 0$ and $\delta = 0.9$ and given their parameters cooperation can be supported at the higher δ . They find much higher cooperation rates at $\delta = 0.9$.

whether cooperation is greater in treatments in which mutual cooperation can be supported in equilibrium.

Table 6 shows cooperation levels depending on whether it can be supported in equilibrium for supergames 1, 7 and 15. The meta-data reveals that indeed, cooperation rates are higher when cooperation can be supported as part of a SPE. Cooperation is significantly greater in average for treatments in which cooperation can be supported in equilibrium in supergame 1. The difference increases importantly as subjects gain experience: by supergame 7 the difference more than doubles the difference under supergame 1. This is due to a decrease in cooperation when it cannot be supported in equilibrium.

Table 6: Equilibrium Condition and Cooperation in Round 1

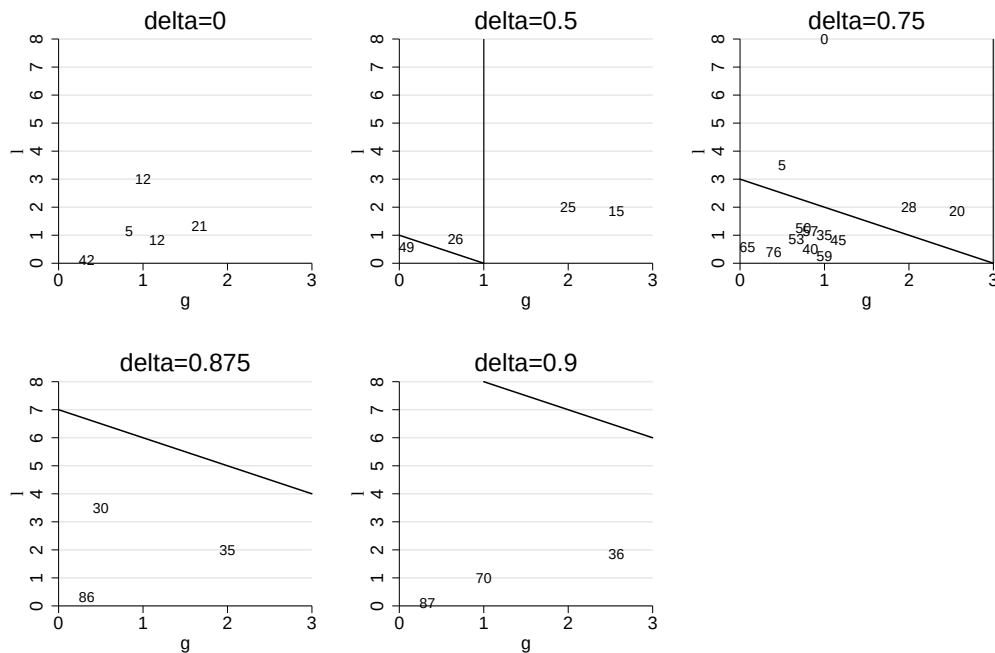
	Not SPE	SPE	Difference
Supergame 1	33.99	50.23	16.24***
Supergame 7	13.25	48.38	35.13***
Supergame 15	16.67	49.67	33.00***

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Figure 3 shows the cooperation percentage by treatment in round 1 of supergame 7. For each δ treatment, the figure also shows a vertical line separating the treatments having cooperation as a SPE outcome to the left, and treatments in which cooperation cannot be supported in equilibrium to the right.¹⁶ As can be seen in Figure 3, while cooperation tends to be higher when it can be supported in equilibrium, there is a lot of variance. Among treatments for which cooperation can be supported in equilibrium, some reach very high levels of cooperation while others reach very low levels of cooperation (compare, for example, the treatment with $\delta = .875$ and $g = \ell = 1/3$ with the treatment with $\delta = .75$ and $g = 1$ and $\ell = 8$). Hence, cooperation being a possible equilibrium action does not necessarily imply that most subjects will coordinate on a cooperative equilibrium. In many of these treatments, a large

¹⁶For example, cooperation can only be supported in equilibrium if $g \leq 1$ when $\delta = 1/2$. Cooperation can be supported in equilibrium for all the available treatments if $\delta = 0.75$ or greater. Cooperation cannot be supported in equilibrium if $\delta = 0$ regardless of g and ℓ .

fraction of subjects chooses to defect instead. This evidence goes against a common practice of assuming that players will make the most of the opportunities to cooperate in equilibrium.¹⁷ Similarly, the experimental literature on coordination games has showed that subjects do not necessarily coordinate on Pareto efficient equilibria (see Van Huyck et al. [1990] and Cooper et al. [1992] among others). Infinitely repeated games are, in a way, a coordination games for high enough δ s. The results show that in infinitely repeated games, subjects do not necessarily coordinate on the best equilibrium as well.



Note: the vertical lines separate the treatments depending on whether cooperation can be supported in a SGP equilibrium and the diagonal lines whether cooperation is risk dominant.

Figure 3: Percentage of Cooperation by Treatment (Supergame 7, Round 1)

A second way to study whether the shadow of the future affects behavior as

¹⁷On this practice, see among many others Tirole [1988] (page 253) who when talking about tacit collusion says: “One natural method (to select from the multiplicity of equilibria) is to assume that the firms coordinate on an equilibrium that yields a Pareto-optimal point in the set of the firms’ equilibrium points.”

theory predicts is to compare infinitely repeated games with finitely repeated games of the same expected length. Under the assumption of monetary payoffs, cooperation should unravel from the end in finitely repeated games. By comparing behavior in the first round of finitely repeated games and infinitely repeated games of the same expected length we can test whether cooperation depends on infinite repetition, as the theory predicts, or it is simply affected by the length of the games.¹⁸ Engle-Warnick and Slonim [2004] compare behavior in finitely repeated and infinitely repeated trust games and, consistent with theory, find that once subjects have gained some experience they trust more and are more trustworthy. Similarly, Dal Bó [2005] finds that cooperation is significantly greater under infinitely repeated PD games than under finitely repeated ones.¹⁹

Result 2 *Cooperation is on average greater in treatments where cooperation can be supported in equilibrium, but there is substantial variation across treatments. The fact that cooperation can be supported in equilibrium does not imply that most subjects will cooperate.*

¹⁸Roth and Murnighan [1978] also mention the importance of such a comparison but they do not conduct finitely repeated PD treatments. Instead, they mention prior evidence that cooperation rates are not affected by the horizon for lengths that overlap with the expected duration of their games.

¹⁹A third way to study whether the “shadow of the future” affects behavior as the theory predict is to exploit variations in the set of possible equilibrium outcomes as a function of the parameters of the game. Dal Bó [2005] exploits the fact that a small variation in payoffs can result in a large variation in the set of outcomes that can be supported in equilibrium. In that paper, one of the payoff matrices allows for mutual cooperation in equilibrium while the other payoff matrix only allows for alternating cooperation in equilibrium when $\delta = 0.5$. Consistent with the theoretical predictions there is more mutual cooperation in the first treatment and some evidence of alternating cooperation in the second treatment. Fréchette and Yuksel [2014] change payoffs in a within-subjects design using $\delta = 0.75$ such that after the payoff change joint cooperation cannot be supported but alternation still can. They observe an important and rapid decrease in cooperation. There are some signs of alternation, but the cooperation rates are so low that it is difficult to evaluate. More work is needed to verify the robustness of these results.

2.3 What Explains Cooperation Levels?

The results from the previous section make it clear that SPE is not enough to understand cooperation levels in infinitely repeated PD games. Then how does cooperation depend on the parameters of the game? When will the multiplicity of equilibria be resolved in favor of cooperation?

The fact that ℓ affects cooperation, as shown in Table 4, casts doubt that satisfying the SPE condition is enough for cooperation to arise (the SPE condition, $\delta \geq \delta^{SPE}$ depends on δ and g only). Why would ℓ affect cooperation? One reason the literature has considered is that subjects may be uncertainty on the choice of the other player. The equilibrium condition (1) is based on the idea that the other player is following the cooperative strategy Grim. However, subjects may not be sure about the strategy of the other player and hence will worry about the cost of cooperating when the other defects: ℓ .

To deal with how strategic uncertainty may affect cooperation in repeated games, the experimental literature has borrowed from the literature on coordination games. The concept of risk dominance (RD) by Harsanyi and Selten [1988] has an easy application in symmetric coordination games with two strategies. A strategy is risk dominant if it is a best response to the other player randomizing 50-50 between the two strategies. While there is an infinite number of strategies in infinitely repeated games, one can focus on two strategies: Always Defect (AD) and a cooperative strategy like Grim (G). We say that cooperation is risk dominant if Grim is risk dominant in the game with only Grim and Always Defect (see Blonski and Spagnolo [2004]).

The condition for cooperation to be part of a risk dominant equilibrium of this simplified game is given by

$$\delta \geq \delta^{RD} = \frac{g + \ell}{1 + g + \ell}. \quad (2)$$

The question of whether risk dominance or subgame perfection determine choices was studied by Dal Bó and Fréchette [2011] and Blonski et al. [2011]. They find that risk dominance does explain cooperation, but that it nonetheless provides an incomplete picture. The same can be said once we consider the meta-data.

Table 7 shows that cooperation rates are higher when cooperation is risk dominant. The difference increases significantly as subjects gain experience: by supergame 7 the difference more than doubles the difference from supergame 1.

Table 7: Risk Dominance and Cooperation in Round 1

	Not RD	RD	Difference
Supergame 1	34.38	54.44	20.06***
Supergame 7	15.10	57.05	41.95***
Supergame 15	19.05	63.11	44.06***

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

However, there are treatments under which cooperation is risk dominant with low levels of cooperation. Figure 3, which shows the cooperation percentage by treatment in round 1 of supergame 7, also shows diagonal lines separating the treatments having cooperation as a risk dominant to the left, and treatments in which cooperation is not risk dominant to the right. Cooperation tends to be higher when it is risk dominant, but there is a lot of differences across treatments (compare for example the treatment with $\delta = .875$, $g = \ell = 1/3$ and the treatment with $\delta = 0.875$, $g = 1/2$ and $\ell = 3.5$). Hence, cooperation being risk dominant does not necessarily imply that most subjects will coordinate on a cooperative equilibrium. In some of these treatments, a large fraction of subjects chooses to defect instead.

Result 3 *Cooperation is on average greater in treatments where cooperation is risk dominant, but there is substantial variation across treatments. The fact that cooperation is risk dominant does not imply that most subjects cooperate.*

This result is quite surprising as it indicates that in infinitely repeated games subjects may not even coordinate on equilibria that are both Pareto efficient and risk dominant.

From the previous analysis we have seen that the concepts of SPE and RD are not enough to explain cooperation levels. There is a lot of variation in

behavior that remains unexplained.

What explains cooperation levels? More precisely, how can the parameters of the game be combined to generate an index that explains cooperation levels? These parameters may not necessarily impact cooperation linearly as assumed in the analysis presented before in Table 4, nor discontinuously following the SPE and RD conditions.

The construction of indexes to explain cooperation in the PD has a long history starting with Roth and Murnighan [1978] for infinitely repeated games and with an even longer history in psychology for PD games in general, see for instance Rapoport and Chammah [1965]. Beside simply considering subgame perfection has a dichotomous predictor for cooperation, some authors have also considered that the magnitude of the incentives to coordinate on cooperation may also matter. Roth and Murnighan [1978] for instance viewed the two indexes, $\frac{g}{1+g}$ and $\frac{1+g}{1+\ell}$ as “measuring the *difficulty* of achieving a cooperative equilibrium.” (p. 196)²⁰

In the same spirit, more recently, Blonski et al. [2011] considered a related index, namely $\delta - \delta^{SPE}$, which measures the difference between the continuation probability and the minimum required for cooperation to be an equilibrium outcome. The first panel of Figure 4 shows the relationship between $\delta - \delta^{SPE}$ and cooperation in the first round of the seventh supergame. This index is positively correlated with cooperation with an apparent increase in slope when $\delta - \delta^{SPE} > 0$. Table 8 presents the results from a probit analysis of the effect of $\delta - \delta^{SPE}$ on cooperation. While for supergame 7 there is no change in slope when $\delta - \delta^{SPE} = 0$, by supergame 15 the slope is not significantly different from zero when $\delta - \delta^{SPE} < 0$ and it is significantly positive when $\delta - \delta^{SPE} > 0$.

There is, however, a significant amount of variation that remains to be explained. In particular this index does poorly for treatments with a high

²⁰The first index comes from the condition for subgame perfection while the second gives the condition for tit-for-tat (TFT) to be an equilibrium. TFT starts by cooperating and after matches what the opponent did in the previous round. There are many more indexes that are constructed by only considering payoffs (but not how they relate to δ) examined in Murnighan and Roth [1983].

value of ℓ as seen by the low levels of cooperation reached in two sessions in Blonski et al. [2011] with high values of ℓ . This lead Blonski et al. [2011] to propose an alternative index: $\delta - \delta^{RD}$. This index attempts to capture how risk dominant cooperation is in a given treatment. The second panel in Figure 4 shows the relationship between this index and cooperation in the first round of the seventh supergame.²¹ Table 8 shows that the effect of $\delta - \delta^{RD}$ on cooperation is positive, especially when cooperation is risk dominant ($\delta - \delta^{RD} > 0$).

While this index exhibits a high correlation with cooperation, it is not clear why the extent to which cooperation is RD for a particular treatment should be measured in the dimension of the discount factor. Notice that the frontier between treatments that have cooperation as RD and treatments that do not is not linear in the δ , g and ℓ space and, hence, ordering the treatments based on the distance to the frontier in the δ dimension may differ from the ordering based on the distance to the frontier in the g or ℓ dimensions.

Dal Bó and Fréchette [2011] address this issue by ordering treatments by how robust they are to strategic uncertainty as captured by the size of the basin of attraction of AD against G: *SizeBAD*. If cooperation can be supported in equilibrium, *SizeBAD* is defined as the maximum probability of the other player following the Grim strategy such that playing Always Defect is optimal.²² This index is equal to $\frac{1}{2}$ if $\delta = \delta^{RD}$, and it is decreasing in δ . The greater *SizeBAD* the weaker cooperation is to strategic uncertainty. For example, if *SizeBAD* = 0.8 a player needs to believe that the other player will cooperate with probability 0.8 or higher for him or her to want to cooperate as well. If cooperation cannot be supported in equilibrium, then *SizeBAD* = 1;

Table 8: The Impact of the Indexes on Cooperation (Round 1 - Marginal Effects)

	Supergame					
	7 (1)	15 (2)	7 (3)	15 (4)	7 (5)	15 (6)
SPE	-0.0831 (0.168)	0.137 (0.150)				
$(\delta - \delta^{SPE}) \times \text{SPE}$	0.732*** (0.107)	0.974*** (0.0718)				
$(\delta - \delta^{SPE}) \times \text{Not SPE}$	0.502* (0.297)	-0.343 (0.280)				
RD			0.152*** (0.0587)	0.157*** (0.0558)	0.307 (0.206)	0.603*** (0.225)
$(\delta - \delta^{RD}) \times \text{RD}$			0.907*** (0.197)	1.496*** (0.318)		
$(\delta - \delta^{RD}) \times \text{Not RD}$			0.212*** (0.0486)	0.147 (0.276)		
<i>SizeBAD</i> × RD					-0.872** (0.345)	-1.191** (0.562)
<i>SizeBAD</i> × Not RD					-0.351* (0.206)	-0.120 (0.422)
<i>N</i>	1984	748	1984	748	1984	748
Different Slope p-value	0.3275	0.0001	0.0004	0.0001	0.2673	0.0662

Clustered Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

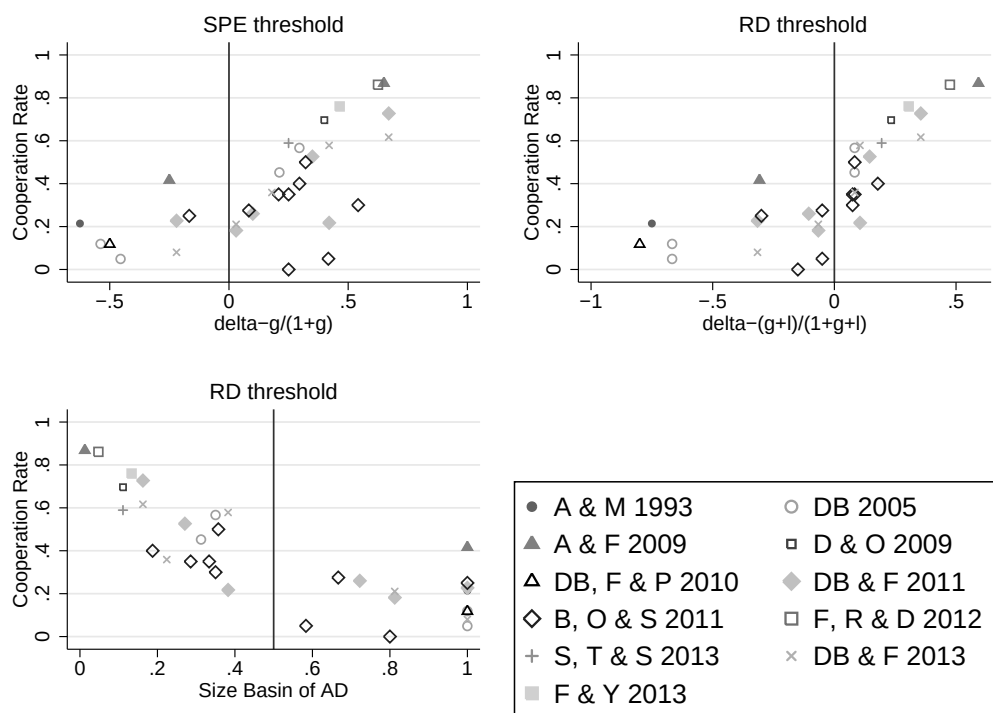


Figure 4: The Relation Between Three Indexes and Cooperation (Round 1, Supergame 7)

AD is optimal regardless of beliefs about the behavior of the other player.²³

Dal Bó and Fréchette [2011] suggest that *SizeBAD* may capture the role of the parameters of the game in determining cooperation. Indeed they show that, in their data, *SizeBAD* correlates with round one cooperation rates in

²¹Rand et al. [2013] presents a figure similar to the second panel in Figure 4 using data from four studies.

²²Formally, *SizeBAD* is defined as follows:

$$SizeBAD = \begin{cases} 1 & \text{if } \delta < \frac{g}{1+g} \\ \frac{(1-\delta)\ell}{(1-(1-\delta)(1+g-\ell))} & \text{otherwise.} \end{cases}$$

²³Myerson [1991] studies the size of the basin of attraction of Always Defect against a cooperative strategy and shows that it is decreasing in the value of δ .

the last supergame of a session. A similar observation can be made using the metadata. The third panel in Figure 4 shows the relation between *SizeBAD* and cooperation rates in round 1 of the seventh supergame. Cooperation rates are relatively low when cooperation cannot be supported by a risk dominant equilibrium but cooperation rates decrease with *SizeBAD* when cooperation is risk dominant. As shown in Table 8, by supergame 15 *SizeBAD* affects cooperation only when cooperation is risk dominant.

Result 4 *Cooperation rates are increasing in how robust cooperation is to strategic uncertainty, especially when cooperation is risk dominant.*

The previous analysis provides a solution to the issue of multiple theoretical predictions. We find that cooperation rates are going to be similar to the ones observed for one-shot PD games when cooperation can be supported in equilibrium but is not risk dominant. Risk dominance does not imply high cooperation rates either. High cooperation rates are only observed for treatments which are very favorable to cooperation (as capture by a small *SizeBAD* or large $\delta - \delta^{RD}$).

An interesting question is which of the three indexes considered above predicts cooperation best. It is clear that $\delta - \delta^{SPE}$ is inferior to $\delta - \delta^{RD}$ and *SizeBAD* as $\delta - \delta^{SPE}$ does not depend on ℓ which has been shown to affect cooperation. Unfortunately, the available data does not allow a robust comparison of $\delta - \delta^{RD}$ and *SizeBAD* given the high correlation of the two indexes in the available data: -0.9158 for supergame 7. This high correlation is due to the choice of treatments from previous papers. Future research should study the explanatory difference between these two indexes by appropriately choosing treatments such that the correlation is not as high. Given that the available data does not allow us to clearly compare the performance of *SizeBAD* and $\delta - \delta^{RD}$, we will focus on *SizeBAD* in future sections due to its better theoretical properties.

2.4 Evolution

The previous section has shown that experience can have large effects on cooperation. Are there factors that mediate experiences in systematic ways? Can

we pinpoint features that affect how behavior evolves? Two such elements have been previously identified: the length of realized supergames and the choices of others.

Engle-Warnick and Slonim [2006a] noted that in infinitely repeated trust games, the length of supergames correlated positively with subsequent choices. Similarly, Dal Bó and Fréchette [2011] show that the length of a supergame affects the likelihood that someone cooperates in the following supergame (even though those are played with different opponents). Dal Bó and Fréchette [2011] also show that a subject who was previously matched with someone who started by cooperating is more likely to cooperate. Both of these observations have been repeated in multiple papers since (Camera and Casari [2009], Sherstyuk et al. [2013], Bernard et al. [2013], Embrey et al. [2013], Fréchette and Yuksel [2014]).

To confirm this, specification (1) of Table 9 reports the marginal effects of a correlated random effects probit of round one choices on: 1) the main factors that were established to be important in the previous section, g , ℓ , δ , dummies for subgame perfection and risk dominance, and *SizeBAD* if cooperation is risk dominant; 2) regressors to account for trends, the supergame number when cooperation is RD and the supergame number when cooperation is not RD; 3) the regressors of interest, the length in rounds of the previous supergame and the round one choice of the opponent in the previous match. The last regressor gives the mean of the random effects which is a function of the choice of the subject in round one of the first supergame. One can think of this as capturing the “type” of the subject. Specification (2) will be discussed later on in this section.

The results clearly indicate that after a longer supergame subjects are more likely to cooperate, and similarly more likely to defect following a short supergame.²⁴ Comparing a situation where the previous match lasted 15 rounds versus one where it lasted 5 rounds increases the probability of cooperation by six percentage points. As a reference, in our sample, supergames have be-

²⁴If instead of the length of the previous supergame we use the deviation from the expected number, results are qualitatively the same. This is not so surprising given that the specification already controls for δ .

Table 9: Cooperation in Round 1

	(1)		(2)	
	Marginal Effects	Standard Errors	Marginal Effects	Standard Errors
g	-0.0058	(0.0129)	-0.0107	(0.0161)
ℓ	-0.0369***	(0.0128)	-0.0325**	(0.0165)
δ	-0.119	(0.0758)	-0.0878	(0.0864)
SPE	0.0914	(0.0668)	0.0846	(0.0668)
RD	0.431***	(0.0694)	0.415***	(0.0747)
$SizeBAD \times RD$	-0.965***	(0.191)	-0.933***	(0.198)
Supergame $\times$ Not RD	-0.0007**	(0.0003)	-0.0007**	(0.0003)
Supergame $\times$ RD	0.0029***	(0.0007)	0.0029***	(0.0007)
Length- E (Length) of Previous Supergame	0.0068***	(0.0009)	0.0068***	(0.0009)
Other's Coop in Previous Supergame	0.120***	(0.0170)	0.120***	(0.0168)
Turnpike			0.0161	(0.0184)
Complete Stranger			-0.0229	(0.0204)
Coop in Supergame 1	0.273***	(0.0391)	0.273***	(0.0389)
N	39,736		39,736	

Clustered Standard errors in parentheses

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

tween one and 44 rounds, with a mean of 2.86 rounds. Does this mean that subjects do not understand how many rounds to expect for a given δ ? Dal Bó [2005] elicited estimates of the expected number of rounds for $\delta = \frac{1}{2}$ and $\frac{3}{4}$ and found that most subjects had in fact a good idea of the expected number of rounds. About 71% and 56% of subjects for $\delta = \frac{1}{2}$ and $\frac{3}{4}$ respectively new the exact expected number of rounds, and the average guess were 1.96 and 3.81 respectively.²⁵ Hence, this suggests that the effect of realized length on choices comes either from the minority of subjects who do not know the expected length, or from subjects updating their overall evaluation of the value of cooperation through experience. One interesting question from the point of view of understanding how humans learn that has not been explored so far is whether or not the impact of the realized length is constant throughout or if the impact is more important early on.

Similarly clear, and non-negligible, is the effect of the choices of the previous player one has interacted with. If the other cooperated in round one of the previous match, a subject is 12 percentage points more likely to cooperate to start his current interaction (which is most probably not with the same person). This process is consistent with subjects having beliefs over the proportion of the population that will cooperate and updating this beliefs as a function of their experiences. As they encounter more subjects that cooperate and update their beliefs upward about the fraction of subjects who cooperate, the value of cooperation increases and thus they are more likely to cooperate themselves. Note that this is in line with the previous observations on the role of the basins of attraction. The role of experience and of the basins of attraction suggest that if beliefs, at the beginning of an experience, are close to the dividing line determined by the *SizeBAD*, the specific experiences could lead to very different long term behavior.

The following result summarizes the above observations.

Result 5 *Cooperation is affected by:*

²⁵Similarly, Murnighan and Roth [1983] asked subjects about the probability that play would continue for at least two more rounds. While subjects tended to overestimate these probabilities, their beliefs strongly depended on δ .

1. *The realized length of previous supergames.*
2. *The choices of the past subjects one was paired with.*

To investigate what would happen in the long run (longer than what can be done in the laboratory), Dal Bó and Fréchette [2011] estimate a belief learning model over supergames. The problem is substantially simplified by focussing on only two strategies: AD and TFT.²⁶ Given this simplification, the supergame can be reduced to the first choice, and learning models that have been previously applied to one-shot games can be used.²⁷ The learning model was estimated subject by subject and the population of estimates was used to simulate sessions. The simulation reproduces the qualitative features of the data over time. This is another piece of evidence that subjects' behavior is consistent with agents who evaluate the value of cooperation based on their beliefs which are affected by their experience and on the parameters of the environment. More interestingly, this allows to simulate what would happen in the very long run (longer than experiments can be conducted with human

²⁶Although an extreme simplification, in that paper these two strategies could perfectly account for behavior in 88% of all matches. This is impressive in itself, but is slightly misleading for two reasons. First, half of the treatments in that paper use $\delta = \frac{1}{2}$, in which case 50% of supergames have only one round. The other half implements $\delta = \frac{3}{4}$ which has 25% of supergames with one round. Hence, for almost 38% of the sample, any two strategies that differ in the first choice will classify all such supergames. Second, in some of these treatments, many supergames result in either both players cooperating all the time or defecting all the time. In those cases, many other strategies beside AD and TFT are consistent with the observations. Thus, the point is not so much the predictive power of these two strategies, but rather that even though this seems like an extreme simplification, it is nonetheless consistent with much of the feedback subjects observe in the laboratory. This being said, it is interesting that even considering the metadata, that includes cases of $\delta = 0.875$ and $\delta = 0.9$, for treatments with $\delta > 0$, 82% of supergames are consistent in every choice with these two strategies. Later in the paper we study the strategies that subjects actually use.

²⁷Ioannou and Romero [2014] propose a method to estimate a learning model while allowing for more strategies. However, their exercise is very different as the learning is not happening between supergames, but rather about learning within supergame. In other words, it describes the considerations of subjects about which choice to make in each round.

subjects). This suggests that even after 1000 matches, there are parameter constellations for which aggregate behavior never coordinates: some subjects cooperate while others defect. In particular, this seems to happen when average cooperation rates start close to the limit of the basin of attraction. One possibility is that in such cases early experiences become more important and lead some subjects to go toward defection while others become more likely to cooperate. However, at this point, the exact process has not been formally established and the role of early experiences is somewhat speculative.

2.5 Strategies

Choices across rounds in infinitely repeated games are likely not independent. In fact, their lack of independence is essential to support cooperation in repeated games: players must condition their behavior on past behavior. Strategies are contingent plans that describe how players condition their behavior on the past.²⁸ Cooperation as an equilibrium outcome or cooperation being part of a risk dominant outcome are concepts that make sense with respect to strategies. Hence, understanding what strategies subjects use is of interest. In addition, the study of the strategies used allows for an additional test of theory. For instance, if subjects are more likely to cooperate when δ increases, but never *punish* defection; then one would want to say that behavior is not consistent with the theory of infinitely repeated games, even though cooperation rates are moving in a way consistent with the predictions of the theory. Identifying strategies could also help predict behavior better, specially after histories that are rarely reached. Finally, it may elucidate aspects of the psychology of humans and how they interact. In other words, characterizing the strategies that subjects use can provide a deeper understanding of behavior.

Two main approaches have been used to study the choice of strategies in infinitely repeated games: the elicitation of strategies and inferring strategies from choices. The elicitation of strategies consists of asking subjects to

²⁸We do not claim that humans form strategies in their head and then execute them, but simply that we can think of their actions as being the outcome of strategies. Strategies here serve as a way to understand the dynamic aspect of how choices interact: both choices that come about as well as those that are never taken given the history of play.

submit a strategy: a plan of behavior for each possible contingency. Eliciting strategies has a long and well known history starting with the work of Robert Axelrod. Axelrod [1980a] reports results for an experiment in which 14 participants (game theorists) submitted strategies for a finitely repeated PD. Axelrod [1980b] considers a randomly repeated PD ($\delta = 0.99654$) and includes 62 entries from game theorists and other participants recruited via announcements in journals for users of small computers. Other papers doing similar tournaments have been written since, but a more common descendent of this method are the computer simulations popular in evolutionary biology.²⁹ The common trait to these papers, be it tournaments or simulations, is that they search for the best performing strategy. From the point of view of studying behavior, however, these can only serve as suggestive evidence since successful strategies may, or may not, be popular.

Here we focus on which strategy people use rather than which strategies perform well. Even though this question is straightforward, answering it via elicitation is marred with difficulties. First, it is not self-evident that people think in terms of strategies (complete contingency plans). Hence, if we ask them, from the onset to formulate a strategy, it might lead them towards behavior that is not what they would have done, had we not triggered them to think about strategies. Second, and again because they may not typically think in terms of strategies, they do not necessarily know how to express what it is they are doing as a strategy. Third, the experiment must allow them to specify the strategy they actually want to use, and to provide a way to evaluate whether the important strategies are included. Dal Bó and Fréchette [2013] propose a method to tackle these potential concerns and elicit strategies for multiple parameters.³⁰

A second way to study the choice of strategies is to infer strategy choice from the round to round choices (cooperate or defect). However, for multiple

²⁹A recent example of a tournament is Selten et al. [1997] who use economics students with some programming experience to program strategies for a finitely repeated asymmetric Cournot duopoly. Examples of computer simulations in evolutionary biology are Nowak and Sigmund [1993] and Nowak et al. [1995].

³⁰Bruttel and Kamecke [2012] partially elicit strategies so as to compare behavior with other ways of implementing infinitely repeated games in the laboratory.

reasons, recovering strategies from choices in infinitely repeated games is a non-trivial problem, even for games as simple as the PD. First, the space of strategies is infinite (uncountable). Second, observed choices are only a small part of a strategy. Third, coordination creates a lack of variation. For example, if all subjects follow the strategy Grim, then we would only observe cooperation in every round and not observe how subjects would react to a defection.

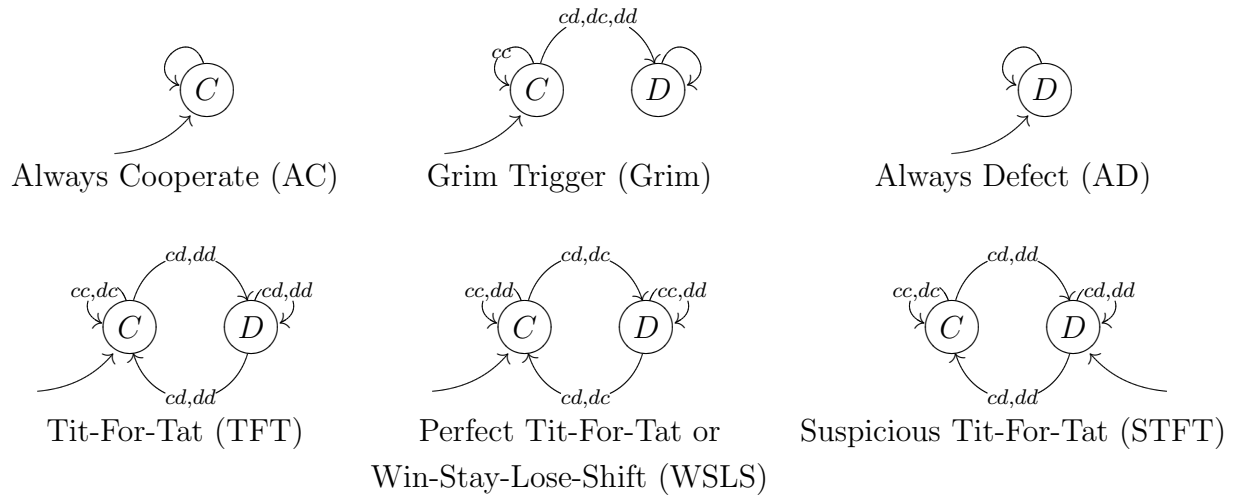


Figure 5: Six Strategies

The various methods that have been proposed to estimate strategies consider the fit of choices and strategies in different ways. While there are many articles with different estimation methods (see Engle-Warnick and Slonim [2006b], Engle-Warnick et al. [2007], Aoyagi and Fréchette [2009], Dal Bó and Fréchette [2011], and Camera et al. [2012]) we will focus on the methodology developed by Dal Bó and Fréchette [2011]: the Strategy Frequency Estimation Method (SFEM). This method consists of estimating a mixture model of the frequency with which each strategy from a pre-specified set of strategies is found in the pooled data under the assumption that each subject always uses the same strategy across supergames but may make mistakes. The estimation consists of choosing the frequency of the different strategies and probability of mistakes that maximize the likelihood of the observed sequences of choices. There are

two possible perspectives on how to interpret these frequencies. One can think of each subject as following the same mixed strategy and the frequencies are the mixing frequencies over the pure strategies.³¹ Alternatively, in the large, these can be fraction of subjects following a particular strategy.³²

There are several reason for us to focus on this identification method. The main reason is that SFEM has been used in several papers over multiple parameters of the PD; allowing us to compare strategy prevalence across payoff matrices, probability of continuation and monitoring structures. Moreover, we think this method has some worth-mentioning advantages over some of the others methods. A very simple advantage is that SFEM allows researchers to use standard statistical methods and tests. Another advantage is that it uses choices from multiple supergames to identify strategies. To see why this is important, consider the following example. For simplicity, assume we are concerned about estimating the strategies of a single subject. Suppose that we have data for many supergames where our subject cooperates in all rounds, but so does his partner. Such data is consistent with many strategies, for instance always cooperate (AC) and Grim can both generate such sequences. However, imagine that in a few supergames, the subject displayed behavior consistent with Grim but not with AC. Then the SFEM would classify all of them as Grim. Other methods can be equivocal, concluding that much of the data is consistent with both AC and Grim.³³

Beside these advantages, there is also evidence that the SFEM performs well (if the relevant strategies are included in the set of possibilities). First, Fudenberg et al. [2012] tested the procedure on simulated data and found that the method produced results consistent with the data generating process. Second, as discussed before, Dal Bó and Fréchette [2013] develop a method to

³¹Note this is mixing before the start of a supergame, from then on the strategy is fixed.

³²In either case, the frequencies should be stable (over supergames). Hence, to minimize the chance that this is not the case, the data set used for estimation is typically composed of supergames late in the session.

³³Note that in certain environments alternative methods may dominate SFEM. Consider for example the environment studied by Aoyagi and Fréchette [2009] with a continuous signal. The SFEM is not practical as it is difficult to know, ex-ante, what are the reasonable thresholds of the signal for triggering punishments.

elicit strategies while also generating choice-by-choice responses. This allows them to compare the results of the SFEM based on the choices of the subjects to the strategies the same subjects selected. The comparison allows them to confirm that estimation results are in line with the strategies subjects selected across a variety of parameters. In addition, it highlights some of the challenges of recovering strategies. In particular, omitting a strategy that is popular can distort results in important ways; especially if the forgotten strategy is “closer” to one that is included than others. In that case, the “closer” strategy will pick up most of what should have been attributed to the missing strategy. Nonetheless, overall the SFEM, and other approaches to estimation, are starting to reveal some clear regularities in the data which we describe below.

Table 10 reports the results of six papers that estimate strategies using the SFEM for infinitely repeated PDs with perfect monitoring. For each treatment of every paper, the estimated percentage of five key strategies is listed. These strategies are represented in Figure 5.³⁴ The most used strategies include AC and AD, which are self explanatory, Grim and TFT, that we have already described, but also Suspicious-Tit-For-Tat (STFT). STFT (sometimes referred to as D-TFT) starts by defecting, and then matches the choice of the other player in the previous round. As can be seen, these five simple strategies account for the majority of strategies in all but one treatment (15 out of 16). Furthermore, in most of these treatments, 13 out of 16, more than three-quarter of the strategies can be accounted for by these five strategies.

Given the work of Axelrod, one could have expected subjects to use TFT. On the other hand, TFT is not subgame perfect, and thus maybe Grim is more

³⁴The representation of strategies as machines works as follows. The machine can be in one of a finite number of states. Transitions between states are governed by conditions. Hence a machine is defined by the list of its states, the triggering condition for each transition, and its initial state. For example, consider the representation of TFT in Figure 5. The arrow coming from outside with no condition indicates the initial state. The letter inside the circle represents the choice the strategy indicates in that state. The letters above the arrows indicate the conditions for transitions which are determined by the pair of actions in the previous round: the first letter for the automaton and the second letter for the opponent’s automaton.

Table 10: Frequency (as %) of Key Strategies

	AC	AD	Grim	TFT	STFT	Total
Dreber et al 2008[§]						
	SFEM Estimation					
$\delta = \frac{3}{4}, g = 2, \ell = 2$	0	64	7	15	14	100
$\delta = \frac{3}{4}, g = 1, \ell = 2$	0	30	21	40	9	100
Dal Bó and Fréchette 2011[§]						
	SFEM Estimation					
$\delta = \frac{1}{2}, g = 2.57, \ell = 1.86$	0	91	0	7	2	100
$\delta = \frac{1}{2}, g = 0.67, \ell = 0.87$	0	76	0	6	8	90
$\delta = \frac{1}{2}, g = 0.09, \ell = 0.57$	1	49	0	24	4	78
$\delta = \frac{3}{4}, g = 2.57, \ell = 1.86$	0	66	0	23	0	89
$\delta = \frac{3}{4}, g = 0.67, \ell = 0.87$	0	11	4	21	8	44
$\delta = \frac{3}{4}, g = 0.09, \ell = 0.57$	2	0	2	55	0	59
Fudenberg, Rand, Dreber 2012[†]						
	SFEM Estimation					
$\delta = \frac{7}{8}, g = 0.33, \ell = 0.33$	24	6	12	15	0	57
Dal Bó and Fréchette 2013[‡]						
	Elicitation					
$\delta = \frac{1}{2}, g = 2.57, \ell = 1.86$	0	53	6	5	14	78
$\delta = \frac{1}{2}, g = 0.09, \ell = 0.57$	3	25	36	19	3	86
$\delta = \frac{3}{4}, g = 2.57, \ell = 1.86$	2	47	10	10	12	81
$\delta = \frac{3}{4}, g = 0.09, \ell = 0.57$	8	12	35	30	0	85
$\delta = \frac{9}{10}, g = 2.57, \ell = 1.86$	2	14	17	39	7	79
Fréchette and Yuksel 2014[†]						
	SFEM Estimation					
$\delta = \frac{3}{4}, g = 0.4, \ell = 0.4$	0	14	32	39	2	87
Rand, Fudenberg, and Dreber 2013[°]						
	SFEM Estimation					
$\delta = \frac{7}{8}, g = 2, \ell = 2$	0	18	43	27	5	93

[§] As reported in Fudenberg et al. [2012], out of 11 strategies.

The original estimation in Dal Bó and Fréchette [2011] included fewer strategies.

[†] Out of 20 strategies.

[‡] Out of 32 strategies or more. The numbers reported are from two treatments, one which allows more than 32 strategies.

[°] Out of 11 strategies.

likely. In fact, both Grim and TFT are present in the data. WSLs (not listed in this table) which is identified in Nowak and Sigmund [1993] as a successful strategy is never a sizeable fraction of the strategies used. AC, which is never an equilibrium, is estimated to have a low prevalence in all treatments. In fact, an even smaller subset of strategies account for most of the behavior: AD, Grim, and TFT. Together, these three strategies cover the majority of strategies in 14 of the 16 treatments and they can account for at least 70% of them in 11 treatments. Hence, under perfect monitoring, the majority of strategies are simple (two-states or less), but not necessarily sub-game perfect equilibria (when playing themselves).

Result 6 *Three strategies account for most of the data: AD, Grim, and TFT. Therefore, subjects do use punishments to support cooperation but punishments are not necessarily credible (not SPE).*

Note that the popularity of AD, Grim, and TFT, seems to extend to more complex PDs with perfect monitoring. In Rand et al. [2013] two treatments are conducted where choices are implemented with error, but this is known to the subjects and they are informed not only of the implemented choices but also of the intended choice of the other player. In Aoyagi et al. [2014] one treatment implements a PD with perfect monitoring where the payoffs are a lottery. AD, Grim, and TFT are estimated to correspond to 64% and 28% in the two Rand et al. [2013] treatments and 61% in Aoyagi et al. [2014].³⁵ Although it is low for one of the three, there are many more possible strategies in this case (since behavior can be conditioned on both what the other player intended to do, as well as the outcome). Together, this is suggestive evidence that these three strategies may be important even in more complex games.

The popularity of simple strategies under perfect monitoring seems to extend to infinitely repeated trust games. Using different estimation methods, Engle-Warnick and Slonim [2006b] find that two-state machines fit more than 80% of the observed behavior (also see Engle-Warnick and Slonim [2004]). Moreover, they find that trustors tend to use the Grim strategy.

³⁵Out of 17 strategies for Rand et al. [2013] and 15 for Aoyagi et al. [2014].

3 Inducing Infinitely Repeated Games in the Laboratory

Before moving on to explore other determinants of cooperation, we take a small methodological detour on inducing infinitely repeated games in the laboratory. The main method to induce infinitely repeated games in the laboratory is to have a random termination rule: after each round the game continues with probability δ and ends with probability $1 - \delta$. Subjects are then paid the sum of payoffs from all rounds. This method was first introduced by Roth and Murnighan [1978] and, under the assumption of risk neutrality, it induces a preference over outcome that coincides with an infinitely repeated game with discount factor δ .³⁶

Several issues may work against the equivalence between infinitely repeated games with exponential discounting and random termination. First, subjects may not understand probabilities and get confused about how the probability of continuation relates to the distribution of supergame lengths. As discussed in section 2.4, the evidence provided by Murnighan and Roth [1983] and Dal Bó [2005] suggests that subjects have a good, but not perfect, perception of how δ affects the distribution of supergame lengths.

Second, subjects may think that the experiment cannot last forever and therefore the constant probability of continuation may not be credible (see Selten et al. [1997] for such a critique). For example, if subjects assign probability zero to the event that the experiment will be still going on next year, then in the last second of the year they will know that they are participating in the last round and cooperation should unravel from the end. Logically, this argument hinges on subjects assigning probability zero to a given experimental length, a year in this example. However, there is no reason to believe that in the last second of the year subjects would still believe that one more round is impossible. Nonetheless, this criticism does lead to an important issue.³⁷

³⁶Note that in theory the model with infinity repetition with exponential discounting is sometimes interpreted as modeling a situation with random termination (see for example Mailath and Samuelson [2006], section 4.) Which interpretation is more appropriate depends on the application.

³⁷From a practical perspective, there are many indications that subjects do not perceive

Many elements outside of the experimenters control may affect the length of a supergame: the server may crash, or there may be an earthquake. As such, it is likely that subjects do not exactly believe that the probability of continuation is the one implemented by the experimenter. Instead subjects may have a subjective perception of the probability of continuation. Therefore, analysis that rely on the exact measure of the probability of continuation need to be taken with a grain of salt and more attention should be paid to analysis that are based on comparisons between treatments (as for example increases in δ , which increase subjects' perception of the probability of future interactions). Of course, this is a comment that applies to experimental work in general.

Third, subjects may not be risk neutral. While it is highly plausible that subjects may display preferences for risk which are not neutral, even with low stakes, the evidence provided in section 4 that risk attitudes do not affect behavior in infinitely repeated games suggests that deviations from risks neutrality are not likely to be a problem. Sherstyuk et al. [2013] also provide evidence in this direction. They compare behavior in random termination PD games when subjects are paid for all rounds and when they are only paid for the last round. They show that both payment methods induce the same preferences if subjects are risk neutral, but different ones if they are not. Consistently with the assumption of risk neutrality, they find that behavior is similar between these two treatments.

Fourth, rematching subjects more than once per session may make it possible to support cooperation even when the parameters are such that in a single randomly terminated game cooperation cannot be supported in equilibrium. Such an argument would rely on some form of contagion type equilibria. Studies exploring the ability of subjects to use such equilibria to

randomly terminated games as they do finitely repeated games. First, there is the evidence from Dal Bó [2005] that cooperation rates are higher in randomly terminated games than in finitely repeated games of the same expected length. Second, in finitely repeated games, Embrey et al. [2014] documents in a meta-study that last round cooperation rates always drop close to zero. There is no similar movement toward no cooperation in randomly terminated games (in cases where cooperation is observed to start with). Third, there is no evidence that cooperation rates drop at the end of sessions implementing randomly terminated games.

support cooperation will be discussed in section 6. More to the point, we can use the meta-data to determine if cooperation rates vary systematically across studies as a function of the matching protocol. The three main protocols used are random rematching (random pairing with replacement), perfect stranger (without replacement), and the turnpike protocol (first introduced in McKelvey and Palfrey [1992]) which eliminates the possibility of contagion. The tradeoff, of course, is that the number of rematches that can be performed is highest for random rematching and lowest for the Turnpike design, and as we have seen, experiencing multiple supergames is important. The regression reported on the right of Table 9 indicates that there are no differences in round one choices across these three rematching designs. Note also that given the large sample size, there are no power issues.

Other types of designs, beside the Roth and Murnighan [1978] implementation, have also been used to induce infinitely repeated games in the laboratory. For example, Sabater-Grande and Georgantzis [2002], Cabral et al. [2014] and Vespa [2014] use a design with a fixed number of rounds which are played with certainty and payoffs are exponentially discounted at rate δ and, after those rounds, a random termination rule is used with probability of continuation δ . Another design, used by Cooper and Kühn [2014a] and Cooper and Kühn [2014b], consists of fixed number of rounds with discounting followed by the coordination game induced by considering two relevant infinitely repeated game strategies like Grim and AD.

Fréchette and Yuksel [2014] compares behavior in experiments with random termination rules (RT), experiments with fixed number of rounds with discounting followed by random termination (D+RT) and fixed rounds with discounting followed by a coordination game (D+C). In addition, Fréchette and Yuksel [2014] introduce a variation on RT in which subjects play in blocks with a fixed number of rounds. Subjects are not informed of whether the supergame has ended or not within a block, and they are only paid for rounds up to termination. A new block of rounds starts only if the supergame has not ended in the previous block. This block random termination design (BRT) allows the researcher to observe behavior in a larger number of rounds than in RT. In addition to the difference in ways to induce infinitely repeated games in the

laboratory, Fréchette and Yuksel [2014] consider two different combination of payoff parameters. In one of these treatments, mutual cooperation can be supported in equilibrium, while in the other treatment it cannot. They find that cooperation is significantly larger in the treatment in which mutual cooperation can be supported in equilibrium under all designs to induce infinitely repeated games. However, the magnitudes of the treatment effects are somewhat different across designs, with the largest effect under RT and the smallest under D+RT.³⁸

In conclusion, there are various different ways to implement infinitely repeated games in the laboratory. Each method presents its specific advantages and disadvantages. The appropriate method will depend on the application one has in mind, on the question being asked, and on which design will best allow one to answer that question. More work needs to be done to understand what differences, if any, different implementation methods generate and what those teach us about how people perceive and react to dynamic incentives.

4 The Impact of Personal Characteristics and Motives on Cooperation

As we have seen, there is a significant amount of heterogeneity of behavior even after subjects have gained experience in a particular treatment. While some subjects attempt to establish cooperative relationships, others defect. Could this heterogeneity in behavior be explained by personal characteristics?

Several articles have explored the connection between risk attitudes and cooperation in infinitely repeated games. Sabater-Grande and Georgantzis [2002] elicit risk preferences by having subjects choose between lotteries and then divide them in three groups based on their risk aversion. Subjects are

³⁸Interestingly, in the D+C design, behavior in the coordination game is largely independent of cooperation in the last round of the PD. This suggests that subjects do not use the coordination game as a way to provide incentives to behave in the PD rounds. Hence, this design is very useful from the point of view of facilitating the study of communication, but it may not be equivalent in the eyes of the subjects to an infinitely repeated game.

matched in each group to play one infinitely repeated game.³⁹ They find that cooperation is negatively related to risk aversion.

Other papers have also studied the relationship between cooperation and risk aversion in addition to other personal characteristics. Dreber et al. [2014] does not find a robust relationship between survey responses to question regarding risk attitudes and cooperation. Similarly, Proto et al. [2014] and Davis et al. [2014] do not find a relationship between risk aversion and cooperation. It is not clear why these three papers found that risk aversion does not correlate with cooperation while Sabater-Grande and Georgantzis [2002] finds that it does.⁴⁰

Another personal characteristic that has been investigated is whether the subject is an Economics major. There is some previous evidence of differences in cooperation rates between economics majors and other students in one-shot prisoners' dilemma games (Frank et al. [1993]). Dal Bó [2005] found that economics majors tend to cooperate less when cooperation is not SPE, but there are no significant differences when cooperation can be supported in equilibrium. However, Dreber et al. [2014] and Sherstyuk et al. [2013] do not find a robust relationship between being an economics major and cooperation.

The literature has also not found a robust relationship between gender and cooperation in infinitely repeated games. While some papers find that women tend to cooperate less in some treatments, other papers do not find such a relationship (see Murnighan and Roth [1983], Dreber et al. [2014], Sherstyuk et al. [2013], Davis et al. [2014] and Proto et al. [2014]).

Note also that besides studying the previously mentioned characteristics, Davis et al. [2014] do not find a correlation between patience and coopera-

³⁹They implement the infinitely repeated game by having 15 rounds with decreasing payoffs (payoffs in round $t + 1$ are 14/15 of the payoffs in round t) and then a probability of continuation of 14/15. Given the payoffs of the PD game, cooperation is both SPE and RD under the assumption of risk neutrality.

⁴⁰One possibility has to do with experience as subjects only participated in one supergame in Sabater-Grande and Georgantzis [2002] while they participated in several in the other papers. A second possibility has to do with the fact that Sabater-Grande and Georgantzis [2002] is the only of the four papers that groups subjects according to the risk aversion. However, subjects were not aware of being grouped by their risk aversion.

tion.

Proto et al. [2014] study the relationship of cooperation to intelligence (among other personal characteristics). Subjects were separated into two groups based on performance in an IQ test and then participated in a series of infinitely repeated PD games. They find that, for a high δ , there are no differences in behavior between the groups in the first supergame, but behavior diverges as they gain experience. Subjects in the high IQ group learn to cooperate. They do not find such a difference for a treatment with a lower δ . They also do not find that personality traits as measured by the big 5 (openness, conscientiousness, extraversion, agreeableness and neuroticism) have a systematic effect on cooperation.

Since appealing to social preferences is common to explain behavior in one-shot games, one would expect that social preferences could also play a role in infinitely repeated games. Dreber et al. [2014] show that there is no correlation between behavior in dictator games and infinitely repeated games in which cooperation can be supported in equilibrium. There is, however, a positive and significant correlation between giving in the dictator game and cooperation in the infinitely repeated prisoners' dilemma game when cooperation cannot be supported in equilibrium. Similarly, Davis et al. [2014] find no correlation between behavior in a trust game and behavior in infinitely repeated games in which cooperation can be supported in equilibrium. In addition, Dreber et al. [2014] survey subjects on their motivations for cooperation and "earning the most points in the long run" was the main motivation for most players. They also show that the type of strategies used by the subjects cannot be better explained by appealing to inequity aversion or altruism. They conclude that social preferences are not important in explaining the heterogeneity of behavior in infinitely repeated games. Behavior is mainly consistent with subjects trying to maximize their monetary payoffs.

Reuben and Suetens [2012] and Cabral et al. [2014] also provide evidence that cooperation in infinitely repeated games is motivated mostly by strategic considerations. By strategic considerations, they mean that subjects' cooperate so that their partner will cooperate in the future. In other words, it is forward looking strategic behavior; as opposed to, for instance, backward

looking reciprocity, or simple altruism. Reuben and Suetens [2012] study an infinitely repeated game in which players can submit their action conditional on whether the round is the last or not. In addition, one of the players can also condition his or her action on the action of the other player. While the parameters are such that cooperation cannot be supported in equilibrium, they find that cooperation is greater when the round is not the last. That is to say that if a subject knows the round is last; he is more likely to defect than if the supergame might continue. This clearly shows that there is a strategic component to cooperation. Cabral et al. [2014] find the same result for infinitely repeated veto games where cooperation can be supported in equilibrium. When subjects condition on the round being last, their choice is more likely to be consistent with a one-shot game than when the supergame might continue. Moreover, Reuben and Suetens [2012] find only a small percentage of cooperation in the last round by subjects informed that the other player cooperates in that round. That is, they find very little intrinsic reciprocity (reciprocating cooperation even when there is no shadow of the future).

Result 7 *There is no robust evidence that risk aversion, economic training, altruism, gender, intelligence, patience, or psychological traits have a systematic effect on cooperation in infinitely repeated games such that cooperation can be supported in equilibrium. There is evidence consistent with the idea that the main motivation behind cooperation is strategic.*

If most cooperation is motivated by strategic considerations, why do we still observe significant levels of defection in treatments which are very favorable to cooperation? Why do some subjects do not realize that it would be highly profitable to start a cooperative relationship?

It is interesting that altruistic and trusting tendencies (as captured by the dictator and trust games) do not seem to play an important role. Future research should continue to search for personal characteristics that would help us predict who will attempt to establish cooperative relationships. It is intriguing that we systematically observe a correlation between the choice in the first round of the first super game and cooperation in later supergames yet few personal characteristic have been found to systematically correlate with be-

havior.⁴¹ Interestingly, Embrey et al. [2013] find that controlling on whether subjects communicated and agreed to cooperate eliminates the correlation between first and subsequent choices suggesting that an important trait might have to do with how subjects react to strategic uncertainty. Hence, a natural question might be: are the subjects who cooperate more likely to coordinate on the Pareto efficient equilibrium in a simple coordination game?

5 On the importance of information

In order to provide incentives for each agent to behave in a manner that is distinct from behavior in a one-shot game, agents must condition future behavior on past actions in a repeated game. Hence, it is no surprise that the ability of agents to monitor the actions of others affects what can be supported in equilibrium. However, even in the absence of perfect monitoring, agents can, in theory, support equilibria with higher expected payoffs than in a static game. What can be supported will depend on the details of the environment and an important factor is the type of monitoring: public or private. In other words, do all agents know what they have all observed (beside the knowledge of their own action) or can their signals be different? This section explores the importance of monitoring on cooperation and is divided in two sections, one for each type of monitoring.

5.1 Imperfect Public Monitoring

Green and Porter [1984] offers the first theoretical analysis of repeated games with imperfect public monitoring. They show that firms playing a Cournot game can support equilibria with lower quantities than the static Cournot

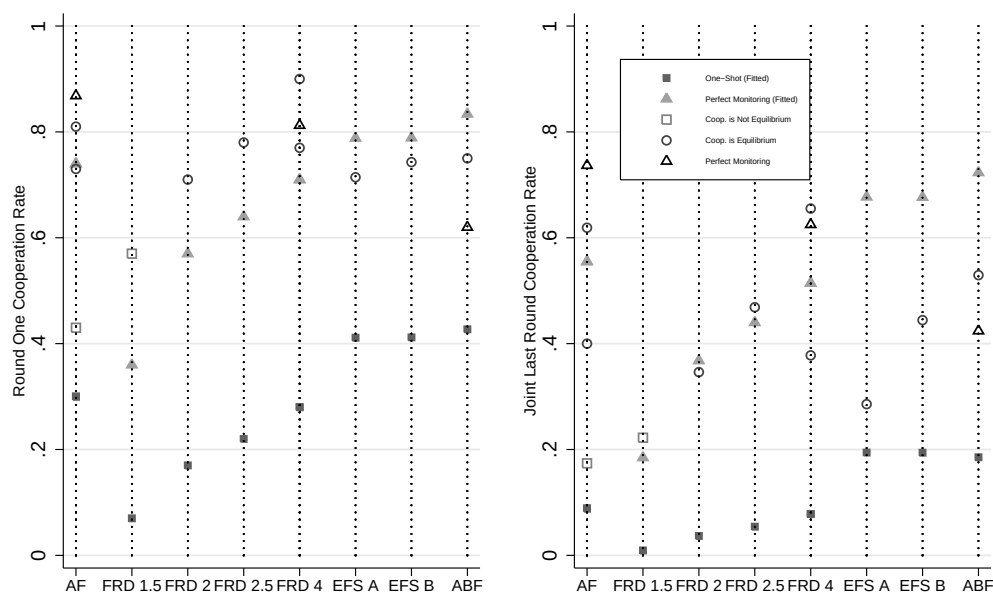
⁴¹The finding that the first choice a subject makes in a session correlates to the subsequent choices in the session is a side-product of estimating a correlated random effects model for cooperation, as we do in Section 2.4. As reported in Table 9, the round one choice correlates with the choices that follow even after controlling for the structural features of the environment (payoffs and discount) as well as other factors. Previous papers have also made this observation not pooling across different treatments (Dal Bó and Fréchette [2011], Embrey et al. [2013], and Fréchette and Yuksel [2014]).

outcome even when there are shocks to demand that make the actions of the players not directly observable. In their setting, firms observe the product price, which is an imperfect (due to the unobservable random shocks to demand) public (they all observe the same price) signal of the players action. They consider trigger strategies where firms produce at the joint monopoly level unless the price in the last period was below a certain *trigger* price, in which case they revert to the Cournot quantity for a certain number of periods. Abreu et al. [1990] characterize the set of Perfect Public Equilibria for given δ under imperfect public monitoring and Fudenberg et al. [1994] provide a Folk Theorem.

This section will cover experimental papers, each with a very different form of imperfect public monitoring. The aim is to understand how imperfect public monitoring affects cooperation and the strategies subjects use to support it.

We will start by discussing the evidence regarding cooperation. Figure 6 shows levels of cooperation observed in several articles studying imperfect public monitoring in infinitely repeated games. As not all papers include treatments with perfect monitoring, we include as a benchmark the levels of cooperation that could be expected under perfect monitoring in the infinitely repeated game based on the analysis from section 2 (this is done as for Table 5). Similarly, we include as another benchmark the levels of cooperation that could be expected from a similar one-shot game. These two benchmarks will help us understand whether subjects are able to reach cooperation levels which are greater than one-shot levels and lower or similar to levels obtained in repeated games with perfect monitoring. This figure presents cooperation levels in round 1 of the seventh supergame in the left panel as a measure of cooperative intentions. In the right panel the figure shows levels of joint cooperation in the last round of the seventh supergame as a measure of the capacity to establish long lasting cooperative relationships (this measure is especially important as punishments happen in equilibrium under imperfect monitoring).

Aoyagi and Fréchette [2009] consider a standard PD in which subjects only receive a noisy public signal of each other's action. The signal is one dimensional and continuous, and hence is similar to the price signal in the Green and Porter [1984] environment. However, subjects play the expected



Hollow markers are data. Solid markers are fitted values from the meta-analysis.
 For the cases with more than one hollow circle, the highest one always had a more precise monitoring.
 AF: Aoyagi and Frechette 2009.
 FRD: Fudenberg et al. 2012, they differentiate their treatments by $(1+g)/l$.
 EFS: Embrey et al. 2013, payoff matrix A and B. Includes structured communication.
 EBF: Aoyagi et al. 2014. Perfect monitoring with random outcomes.

Figure 6: Cooperation under Imperfect Public Monitoring (Supergame 7)

value game and thus their signal does not affect realized payoffs.⁴² The paper analytically derives the maximal symmetric perfect public equilibrium payoff for a specific distribution of the random component of the public signal and the authors show that it is decreasing in its variance (the level of noise). The experiment shows that indeed, subject can support cooperation even when monitoring is imperfect. As seen in Figure 6, when it can be supported in equilibrium, round 1 cooperation under imperfect public moni-

⁴²At the end of each supergame, subjects are informed of their payoffs. If they did not play the expected value of the game, subjects could infer choices from the payoffs and the public signal. Note that, in theory, the way noise affects payoffs should not matter, only the strategic form of the game is relevant. However, it could be that this does matter for how people behave. The evidence so far is that it does not (see footnote 56 for more details), but in the interest of completeness, we will be precise about how noise determines payoffs in the way each experiment is implemented.

toring in Aoyagi and Fréchette [2009] is similar to cooperation under perfect information. This does not mean that information does not matter: in line with the maximal payoff, behavior is such that as monitoring becomes noisier, cooperation and average payoffs decrease.

Fudenberg et al. [2012] implement a very different public signal: it is finite, two dimensional, and affects payoffs. More specifically, after making their choice in the PD, there is some probability ($\frac{1}{8}$ in most treatments) that each player's choice will be altered (to be the other choice). The modified choices are the ones that determine payoffs and both players are informed of the implemented (modified) choices.⁴³ Figure 6 shows the results for each of the four payoff matrices considered by Fudenberg et al. [2012]. They find that subjects can support cooperation despite imperfect monitoring. For each of the payoff matrices, the figure shows that cooperation under imperfect public monitoring is greater or comparable to what we could expect under perfect monitoring.

Embrey et al. [2013] study imperfect public monitoring in a repeated partnership game. In the stage game, subjects simultaneously choose a costly effort level to contribute to a joint project. The higher the chosen effort levels, the higher the chance the project will be successful. At the end of the stage, subjects do not observe the effort level chosen by their partner; instead, they only observe whether the project was a failure or a success, with the latter resulting in both subjects receiving a higher payoff. They consider versions of the game with two or three available effort levels. When only two effort levels are available, the game is a PD in expected value (they consider two different payoff matrices: A and B). In this case, the imperfect monitoring environment has a unidimensional, binary – and hence very coarse – public signal that has direct payoff consequences. Again, the results indicate that subjects can support cooperation despite imperfect monitoring, as seen in Figure 6. Inter-

⁴³Note that one important difference between a setting such as this one and the one, say, of Aoyagi and Fréchette [2009], is that in the former, deviations can be statistically identified while in the later they cannot. When deviations can be statistically identified, there is common knowledge of who's "choice" was defect (even if that was not their actual choice). This could affect the choice of strategies.

estingly, joint cooperation levels in the last round are much lower, consistent with the idea that punishments will be triggered by bad signals.

Using this framework, Embrey et al. [2013] explore the role of renegotiation proofness in equilibrium selection, a selection concept that has drawn a great deal of attention in the theoretical literature.⁴⁴ The requirement of renegotiation proofness reduces how much cooperation can be supported in equilibrium as it limits the severity of the punishment phases; more severe punishments are no longer credible as they would be renegotiated should they be reached. Loosely speaking, the frequent observation that payoffs in prior experiments are below the maximum equilibrium payoffs could be because the harsh punishments required to sustain high levels of cooperation are not credible due to renegotiation concerns. Results from this experiment, however, suggest that subjects do not select the renegotiation proof strategy. By adding a medium effort level to the two-choice stage game, the maximum cooperation that can be supported in a renegotiation-proof equilibrium decreases from the high to the medium effort level. Despite this prediction, subjects rarely select the medium effort level. On the other hand, in line with the renegotiation proofness logic, when the changes in stage games prescribe the use of short (forgiving) punishment phases, amongst cooperative strategies, forgiving strategies are indeed more common.

Aoyagi et al. [2014] explore yet another implementation of the PD with imperfect public monitoring.⁴⁵ The signal an agent receives of the other's choice corresponds to the correct choice with probability 0.9, but with 0.1 chance it is the opposite choice. Payoffs are determined by one's own choice and the signal he receives of the other's choice. Both players are informed of the signal the other received of his own choice. Hence, as in Fudenberg et al. [2012], the public signal is finite, two dimensional, and affects payoffs. However, unlike in Fudenberg et al. [2012], for example if both players cooperate but the other

⁴⁴Unlike most papers in this literature, Embrey et al. [2013] include a structured, non-binding communication protocol through which agents can agree to an action combination or to a conditional action plan – of the type, do this now then if the project is a success we do this, if the project is a failure we do that.

⁴⁵This paper focuses on private monitoring and thus will be described in more details in the next subsection.

has been told they defected, their payoff would both be the *sucker's* payoff, while it would be the *punishment* payoff in Fudenberg et al. [2012]. Surprisingly, as see in Figure 6, round one cooperation in supergame seven is greater under imperfect public monitoring than under perfect monitoring (but this difference is not statistically significant).

The finding in Aoyagi and Fréchette [2009] that increased noise leads to lower payoffs seems specific to their environment. Both Fudenberg et al. [2012] and Aoyagi et al. [2014] who have treatments with perfect and imperfect public monitoring do not observe lower payoffs under imperfect monitoring. It may be that the relation between monitoring and payoffs depends on the form of the public signal.

Finally, Rojas [2012] uses a clever design to implement imperfect public monitoring.⁴⁶ Subjects play a randomly terminated PD where the specific payoffs can be one of three possibilities corresponding to low, medium, or high demand in a Cournot game.⁴⁷ After each round, subjects are informed of their payoffs. However, some payoff numbers are the same for different combinations of the other player's choice and demand. For instance, the payoff to seller one when he selects the *High* output is the same if seller two selects the *Low* output and demand is medium or if seller two selects the *High* output and demand is high. In the perfect monitoring treatment, after choices are made, subjects are informed of the state of demand that prevailed, hence they can infer their opponent's choice from their payoff. In the imperfect public monitoring condition, subjects are not informed of the demand state. Given the structure, subjects can reconstruct what must be the prevailing prices (to rationalize profits) and those prices form an imperfect but public signal.⁴⁸ The experiment uses a between subjects design and four treatments

⁴⁶We describe only two of the three treatments implemented here. The third treatment will be mentioned later in the survey.

⁴⁷Unlike most studies, instructions are given *in context*, using terms such as sellers, market, demand, and output.

⁴⁸We note that this mapping is non-trivial and that if subjects do not or cannot perform it, the environment may be perceived as one of private monitoring. Furthermore, although this monitoring structure is public, it has two features that are not standard. First, the public signal is not always common knowledge. Second, the support of the public signal

are implemented: one payoff structure with δ of 0.6 or 0.75, and another payoff structure which is more conducive to cooperation with δ equal to 0.75 and 0.9. The author shows that using Grim and assuming risk neutrality, cooperation can be supported in all four conditions with perfect monitoring, but only with the second payoff specification with imperfect monitoring. Across all four treatments, cooperation rates vary between 0.21 and 0.45 under perfect monitoring and between 0.24 and 0.34 under imperfect monitoring. In fact, two treatments find a small increase in cooperation going from perfect to imperfect monitoring. Furthermore, the trend over supergames is either flat or slightly decreasing in three of the four treatments. Overall, the ability of subjects to support cooperation in this environment seems modest even with perfect monitoring and the impact of imperfect monitoring seems unclear.

Result 8 *Subjects attempt to cooperate even with imperfect public information about their partner's behavior. However, in some of these environments, cooperation is difficult to sustain.*

The literature has not only paid attention to the levels of cooperation reached under imperfect public monitoring, but has also studied the strategies used by subjects in these environments. Notice a key difference from the case with perfect monitoring considered so far: under imperfect monitoring agents must sometimes enter a punishment phase on the equilibrium path, even though they understand nobody is “cheating,” otherwise there are no incentives to cooperate. This difference could be meaningful when it comes to analyzing strategies. In standard perfect monitoring PD games, it is much easier to identify a player's strategy when he is matched with other players that use varying strategies or use strategies that have random changes. This allows us to see how the player responds to these different situations. In an experiment, observing a variety of strategies that are somewhat *random* is more likely to happen early on. However, the behavior of interest is the one towards the end, when subjects have as much experience as possible. Unfortunately, as behavior stabilizes and subjects coordinate, variability is decreased, making it more difficult to observe the contingent play that would allow us to

varies with choices.

infer the strategies being used. This is not the case with imperfect monitoring since punishment phases are observed in equilibrium. This makes imperfect monitoring environments potentially more revealing about strategies. On the other hand, it could be the case that the strategies used are sensitive to the monitoring technology, something that theory predicts.

Several articles have studied strategies under imperfect public monitoring. First, Aoyagi and Fréchette [2009] estimate that the strategy most commonly used is a two-state threshold strategy which is forgiving (it can come back to cooperation after a defection) and that the threshold to move to a punishment phase is more lenient than what the theory predicts to support the maximum symmetric equilibrium payoff.

One contribution of Fudenberg et al. [2012] that has proven to be very important is to identify that subjects tend to use strategies that are lenient (do not immediately trigger a punishment) or forgiving (come back to cooperation after a punishment phase) in the environment they study.⁴⁹ These includes versions of Grim that delay punishments until two or three consecutive bad signals have been observed. It also includes variations on TFT that delay triggering a punishment or require more than one good signal to return to cooperation after a punishment has been triggered. This can be seen in Table 11 that summarizes information from perfect monitoring treatments already shown in Table 10 and includes data from studies with imperfect public monitoring.⁵⁰ The results for Fudenberg et al. [2012] are separated between the one treatment they have with perfect monitoring and the treatments with imperfect public monitoring. One can obtain almost all of the lenient or forgiving strategies that start by cooperating by adding the frequency of TFT and of strategies with more than two states that start by cooperating, abbreviated

⁴⁹Note that leniency and forgiveness in a finite signal environment must take a very different form from that in a continuous signal environment.

⁵⁰The table separately gives the percentages for AD, Grim, and TFT, which were identified as the most common strategies under perfect monitoring. Note that TFT is a forgiving strategy. “> 2 states (C first)” consists only of lenient and forgiving strategies that start by cooperating. “2 states (C first)” includes some forgiving strategies, such as WSLs, and some that are neither lenient nor forgiving, like the strategy that starts with cooperation followed by defection forever.

as “> 2 states (C first)”.⁵¹ The striking result from that paper can be observed by noting that TFT and of “> 2 states (C first)” go from fitting 47% of their data when monitoring is perfect, to 74% of the data, when monitoring is imperfect. Unlike Aoyagi and Fréchette [2009], the results points to many strategies with more than two states.

Looking at Table 11 we see that these papers (Dreber et al. [2008], Dal Bó and Fréchette [2011], Fréchette and Yuksel [2014], Rand et al. [2013], Embrey et al. [2013], and Aoyagi et al. [2014]) together confirm the finding of Fudenberg et al. [2012]. Namely, as monitoring becomes imperfect, a non-negligible fraction of subjects migrate towards forgiving or lenient strategies. Noticeably, the fraction of strategies accounted for by “> 2 states (C first)”, which are all lenient and forgiving, tends to be higher under imperfect public monitoring than under perfect monitoring (51% versus 14.6% averaging over studies). Note that, theoretically, there is no need for this change: the Grim strategy can support cooperation in all of these treatments. However, the Grim strategy leads to potentially non-negligible efficiency loss in noisy environments. Hence, the change in strategies could be to mitigate the efficiency cost of non-lenient/non-forgiving strategies in noisy environments. We also note that TFT, which is forgiving, does not seem to increase in popularity under imperfect public monitoring, if anything it decreases in popularity. Hence, this movement towards leniency and forgiveness also accompanies a movement towards more complexity.

The finding of increased leniency and forgiveness does not seem constrained to discrete signals. Aoyagi and Fréchette [2009] report that as noise increase in their experiment, the median threshold to go from cooperation to defection and from defection to cooperation decreases, which in their case means that strategies are becoming more lenient (they require a worse signal to trigger a punishment) and more forgiving (they do not require as good a signal to return to cooperation). Hence, in their setting, leniency and forgiveness is easily changed without changing the complexity of the strategy used.

⁵¹This excludes two forgiving strategies, WSLS, and a strategy that punishes a bad signal by a single period of punishment and then returns to cooperation. With the exception of one case, neither of these strategies are popular.

Table 11: Percentage of Strategy Types in Perfect and Imperfect Public Monitoring

Monitoring:	Perfect					Imperfect Public		
	Dreber et al 2008 [§]	Dal Bó and Fréchette 2011 [§]	Fudenberg et al 2012	Fréchette and Yuksel 2014	Rand et al 2013	Fudenberg et al 2012	Embrey et al 2013 [†]	Aoyagi et al 2014
AD	47	40	6	14	18	14	18	19
Grim	14	5	12	34	43	5	6	4
TFT	28	26	15	36	27	7	11	0
AD+Grim+TFT	89	71	33	84	88	26	35	23
Other 2 states (C first)			2	1		2	10	3
> 2 states (C first)	0	24	32	10	7	67	33	53
2 states (D first)	12	4	0	6	5	2	0	0
> 2 states (D first)			8	0		2	14	3
AC	0	1	24	0	0	2	12	20

Results based on SFEM estimation (restricted to studies that included a large set of strategies).

Only includes treatments where joint cooperation can be supported in equilibrium.

Average over treatments (not weighted) if there are more than one treatment.

Entries left blank were not considered in those specific estimations.

[§] As reported in Fudenberg et al (2012).

[†] Only includes stage games with two choices.

Result 9 *Imperfect public monitoring results in subjects using more forgiving and lenient strategies relative to perfect monitoring.*

5.2 Imperfect Private Monitoring

Many economic situations involve repeated interactions where players do not know for sure what the other players did, nor do they know exactly what the other players have observed. A classic example of such imperfect monitoring is taken from Stigler [1964], in which he describes the difficulty of colluding in an oligopoly where firms post prices but can offer secret price discounts. These discounts are not observed by other firms. Instead, each firm receives a private signal of the others' behavior through its sales – sales are an imperfect signal due to random shocks to demand.

Although examples where private monitoring is present seem ubiquitous, theory has only recently made progress on this front. In fact, Kandori [2002] wrote about what players can achieve in repeated games with private monitoring: “This is probably one of the best known long-standing open questions in economic theory.” (p. 3) The absence of a commonly observed signal introduces a significant problem: how should agents use histories to coordinate the continuation play to provide the correct incentives given that they do not know each other's history?

Finding testable implications of the theory of infinitely repeated games can be challenging, but the challenge is even greater with imperfect private monitoring. For instance, establishing whether a specific strategy profile can be supported in equilibrium is often difficult. General results such as the highest symmetric equilibrium payoff are still unknown. For these reasons it is even more important for experiments on this topic to put behavior in relation to something else to make sense of it (cross treatment comparative statics or a theoretical benchmark).

Without going into details, let us highlight some of the well known results from this literature (see Mailath and Samuelson [2006] for a survey). One important difference between private and public monitoring is that no pure strategy profile that supports cooperation can be an equilibrium under im-

perfect private monitoring. Hence, any equilibrium that supports cooperation will require mixing. Two main approaches are used: the belief based and the belief free approaches.⁵² Under the belief based approach, agents mix over simple strategies (for instance Grim and AD) such that both players make each other indifferent between the two strategies and punishments are credible (see Sekiguchi [1997] and Bhaskar and Van Damme [2002]). Under the belief free approach, the strategies are constructed such that in every round subjects are indifferent between cooperation and defection (Piccione [2002] and Ely and Välimäki [2002]).

Not surprisingly given the challenges highlighted above, evidence from experiments with imperfect private monitoring is scant, with only three papers to date exploring private monitoring with randomly terminated games. The first paper that falls in this category is Feinberg and Snyder [2002] where they explore a modified PD with three choices (the third action gives both players the *sucker's* payoff independent of the other's choice). Both sessions were composed of two supergames.⁵³ In the first supergame, which is played with perfect monitoring, there were eight rounds played for sure and after that each additional round occurs with probability $\frac{2}{3}$. The second supergame has 18 rounds for sure followed by the same random termination as the first supergame. In that second supergame, imperfect monitoring is introduced by occasionally manipulating the payoff numbers (giving the *sucker's* payoff to both players), and this is known to subjects. In one treatment (one session) there is ex post revelation of those manipulations and in the other there is not. Importantly, in the second supergame, subjects are only informed of their own payoff, they do not know the payoff of the person they are playing with. In the case where subjects are not informed of the payoff manipulations, this introduces private monitoring since when a subject receives the *sucker's* payoff, he does not know what his partner did nor what signal his partner received. They report that subjects cooperate 62% of the time in the first supergame with no shocks. In the second supergame with ex-post revelation of payoff manipulations, subjects cooperate in 68% of their choices. However, when they are not

⁵²See also Compte and Postlewaite [2010] for an alternative *behavioral* approach.

⁵³In total only 16 subjects participated in this experiment.

informed of those manipulations, they cooperate at a rate of only 21%. Hence, private monitoring severely hampered the ability of subjects to cooperate in this environment. Note, however, that one difficulty in analyzing these results is that we do not know if cooperation can be supported theoretically. Hence, whether or not a drop of this magnitude should or should not be expected is unclear.

Matsushima and Toyama [2011] conduct an experiment with an environment closer to most of the standard theoretical work on private monitoring. Subjects play a PD and receive an imperfect and private signal of what the other player selected. Their payoffs are determined by the actual choices (in this aspect it is closer to Aoyagi and Fréchette [2009] than other designs). In all sessions, $g = \ell = 0.2$ and $\delta = 0.967$, making it the treatment with the highest δ conducted so far. The treatment variable is the accuracy of the signal, either the probability that the signal corresponds to the choice is 0.9 (high accuracy) or 0.6 (low accuracy). The design is a mix of within and between subjects design: two sessions of three supergames with high accuracy followed by three supergames with low accuracy and two sessions with the reverse order. The authors use two elements to understand the data: the comparison of behavior across the two treatments and the comparison of the data to the implications of a specific equilibrium (symmetric generous tit-for-tat; which is the best symmetric belief free equilibrium with memory-one). The symmetric generous tit-for-tat (SGTFT) cooperates with some probability in each round, the specific probability is determined by the round (round one versus all other rounds) and the signal of the other's choice in the previous round (for rounds other than one). If subjects select this equilibrium in both treatments, then an implication of the theory is that in the low accuracy treatment the difference in cooperation rates between the case where the last signal was *good* versus when it was *bad* should be greater than in the high accuracy treatment. The intuition is that as the monitoring becomes worst, it is more difficult to incentivise agents and thus more intensity in reaction is required. The key results are the following: 1) cooperation rates are high in both treatments; 2) cooperation rates are higher when accuracy is higher; 3) cooperation rates are lower than predicted by the SGTFT and 4) the reaction intensity to the last round's

signal is *higher* in the high accuracy treatment. Some of the difficulties in interpreting these results can be understood taking into consideration what has been established in the previous sections. The payoff parameters used in this study are very conducive to cooperation. Using the results of the meta study to predict cooperation rates in a one-shot game with those payoffs over the course of six supergames, we find a predicted average of 44.4%.⁵⁴ The round one cooperation rate in their data for the low accuracy treatment is 43.8%. This indirectly indicates that subjects are actually not cooperating more in that treatment than they would in a one-shot game. Relatedly, the intensity of reaction in that treatment is only 0.165, which although it is not reported, may not be statistically different from zero. On the other hand, in the high accuracy treatment, cooperation in round one averages 78.1% and the intensity of reactions is 0.508. This certainly is in line with subjects supporting cooperation using some strategies that respond to the private signal. It does suggest also that the strategies played are different in both conditions, and thus interpretations about reciprocity deduced from the comparison to the theoretical implications of SGTFT may be misleading. Furthermore, as we have seen in Section 5.1 (and first proposed by Fudenberg et al. [2012]), when moving to imperfect public monitoring, many of the strategies used depend on more than the signal just from the previous round. If this finding extends to private monitoring, an analysis that relies exclusively on memory-one strategies would give the wrong benchmark.

Aoyagi et al. [2014] use a different approach to interpret behavior under imperfect private monitoring. Instead of comparing behavior to a particular equilibrium, it compares behavior across various monitoring environments: perfect, imperfect public, and imperfect private. In order to do this, they use the following design. Under imperfect monitoring, both public and private, subjects choose an action in a PD, and with some probability the signal of their choice that is communicated to the other player is incorrect. Their payoff is determined by their own choice and the signal they receive of what the other did. The difference between the public and private treatment is that under public monitoring both players know not only their choice and

⁵⁴The paper does not report what happens over time in different supergames.

the signal of what the other did, as under private monitoring, but they also know the signal the other received of their own choice. In the perfect monitoring treatment subjects know exactly the choice of both players. However, the payoffs are determined by lotteries equivalent to the lottery faced by a subject under imperfect monitoring.⁵⁵ Hence, this design keeps the strategic form of the game fixed across treatments, i.e. expected payoffs as a function of actions are the same for perfect, imperfect public, and imperfect private monitoring.⁵⁶ The parameters are: $g = \ell \approx 0.357$, $\delta = 0.9$, and the probability that the signal is incorrect is 0.1.⁵⁷ Under these parameters, theoretically, (some) cooperation can be supported in all three treatments. The key results with respect to private monitoring are the following. Cooperation can be supported under private monitoring. In addition, this happens at rates very similar, albeit slightly lower than, under perfect and public monitoring. The rates of round one cooperation in the last four supergames are 65% and 73% for perfect and public, while they are 61% for private. This confirms the result of Matsushima and Toyama [2011] that cooperation can be supported under private monitoring, but it extends it to a different environment and establishes that this can be done at rates similar to what happens under other monitoring structures. In addition, the paper shows that, in line with the intuition from the theory, coordination (of actions) is higher under perfect and public monitoring than it is under private monitoring. Finally, as al-

⁵⁵Under imperfect monitoring, for example, if two players cooperate, they each face a lottery that with some probability they receive the *reward* payoff, unless the signal they receive of the other's action is incorrect (which happens with the complementary probability) in which case they receive the *sucker's* payoff.

⁵⁶ Note that behavior in the perfect monitoring treatment suggests that the outcome of the lottery does not have an important effect on choices, and, if anything, the effect diminishes with experience. Rand et al. [2014] directly studies this question, namely it compares behavior in an imperfect public monitoring treatment (with the same structure as Fudenberg et al. [2012]) to a similar treatment where both the implemented choice and intended choice are revealed to both players (hence this treatment has perfect monitoring). Although they document some differences between the two treatments, the results overall point to most subjects reacting to intentions, not to outcomes.

⁵⁷Depending on the session (there are four sessions per treatment), as few as nine and as many as 22 supergames are completed.

ready mentioned, Fudenberg et al. [2012] established the popularity of lenient and forgiving strategies under public monitoring and we confirmed that across multiple studies subjects use more lenient and forgiving strategies to support cooperation under public monitoring than they do under perfect monitoring. A related movement towards more complex strategies is observed here as well under private monitoring. However, what is new is the observation that under private monitoring, strategies are as lenient as they are under public monitoring, but in terms of forgiveness, they are similar to what is observed under perfect monitoring. In particular, lenient forms of grim that wait for two or three consecutive bad signals before triggering a punishment phase are significantly more popular under private monitoring than under either perfect or public monitoring.

While more research is needed on imperfect private monitoring, the existent work leads to the following conclusion.

Result 10 *Cooperation is possible under imperfect private monitoring and subjects seem to move towards adopting more lenient strategies.*

6 Community Enforcement

Previously we have considered situations in which the same set of players interacts over and over. However, in many economically important situations, the same set of players do not always interact together but switch partners over time. For example, consider how the matching of consumers to firms or the compositions of teams in firms may change from time to time. Many market and other types of interactions are characterized with free riding opportunities arising from limiting contracting or asymmetric information. Overcoming these opportunities is essential for markets and societies to work efficiently. While this may seem impossible when the changing of partners eliminates the possibility for personal retaliation, a fascinating theoretical literature has shown that cooperation may still be a possible equilibrium outcome (see Kandori [1992], Greif [1993], Ellison [1994] and Okuno-Fujiwara and Postlewaite [1995]).

Opportunistic behavior can be overcome even with changing partners as groups may follow social norms in which opportunistic behavior is punished by members of the group, even by those who did not suffer the cost of that opportunistic behavior. Does community enforcement lead to cooperation in repeated games with changing partners? A series of recent papers have addressed this question under a variety of environments.

One of the environments studied by the experimental literature on community enforcement is random matching with anonymous players and personal histories (Schwartz et al. [2000], Duffy and Ochs [2009] and Camera and Casari [2009]). Anonymity refers to the fact that players do not learn the identity of their partners. As such it is impossible to specifically retaliate against a defector by identifying him or her (there is no possibility of establishing a relationship based on personal enforcement as in the previous sections). Personal histories refers to the fact that subjects observe what their partners do but have no information of what happened in other interactions. Contrary to what one would initially think, Kandori [1992] showed that it may still be possible to support cooperation through community enforcement when the group is small enough (the size of the group that can allow for cooperation in equilibrium depends on the PD payoffs parameters and discount factor). While players do not observe what happened in other pairs nor know the identity of their partners, they can follow a “contagion” strategy that reverts to defection after observing a defection. If players follow this strategy, an initial defection will result in a community wide collapse of cooperation and the initial defector will be punished even when no member of the society can identify who the initial defector was.

Duffy and Ochs [2009] study behavior in infinitely repeated games with anonymous random matching and personal histories with groups of 6 and 14 subjects. Subjects are randomly matched in pairs in each round to play a prisoners’ dilemma game with $g = \ell = 1$ and probability of continuation $\delta = 0.9$. In this treatments cooperation can be supported in equilibrium following Kandori [1992]. They also run sessions in which subjects are matched in fixed pairs for the whole supergame.⁵⁸

⁵⁸This is the treatment that is included in the data described in Table 3.

Figure 7 shows the evolution of first round cooperation levels by supergame for both treatments with random matching (groups of 6 and groups of 14) and for the fixed pairs treatment. In addition, we include as benchmarks the cooperation levels that we would expect for fixed pairs for $\delta = 0.9$ and for one-shot games (this is done as for Table 5 in section 2).

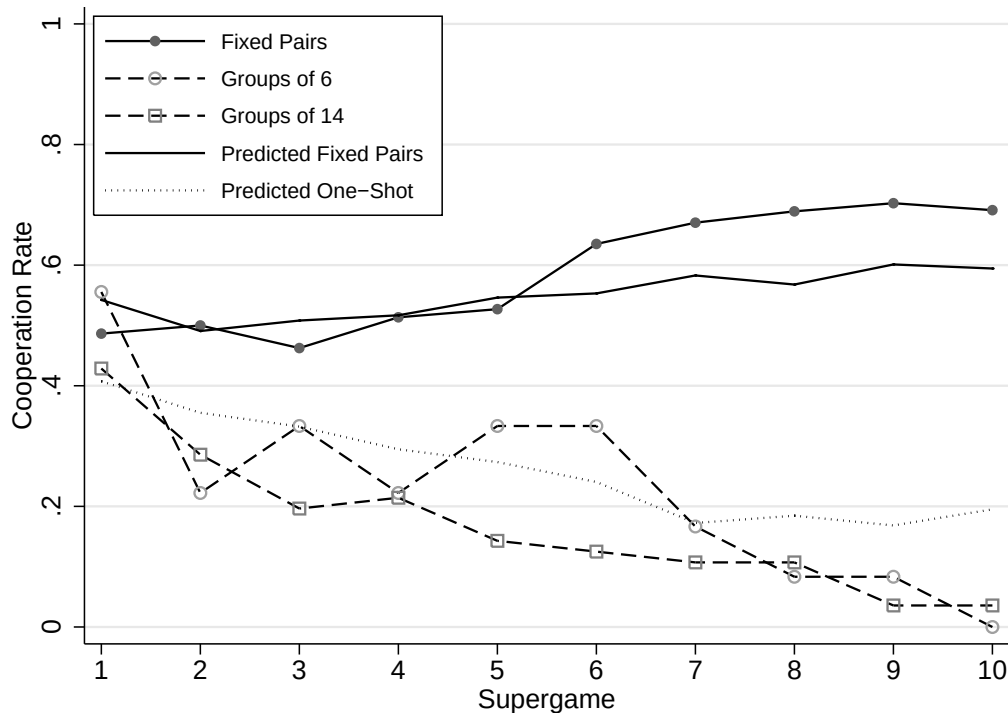


Figure 7: Round 1 Cooperation in Anonymous Random Matching Games with Personal Histories - Data from Duffy and Ochs [2009]

The evolution of cooperation under fixed pairs is consistent with what could be expected based on the analysis in section 2. Cooperation increases slightly with experience. For the treatments with random matching, however, cooperation does what one would expect in one-shot games. After even just a little experience, and regardless of the size of the group, random matching leads to levels of cooperation that are quite different from those from repeated interaction under fixed pairs. In conclusion, the results from Duffy and Ochs

[2009] suggests that subjects may not use the opportunity to cooperate that arises from the contagion equilibrium identified in Kandori [1992].⁵⁹

Camera and Casari [2009] study behavior in infinitely repeated games with anonymous random matching and personal histories with groups of 4 and 14 subjects. In each period, subjects are randomly matched in pairs to play a prisoners' dilemma game with $g = \ell = 1/3$ and probability of continuation $\delta = 0.95$. Figure 8 shows first round cooperation levels by supergame for both treatments (groups of 4 and groups of 14). We also include as benchmarks the cooperation levels that we would expect for fixed pairs for $\delta = 0.95$ and for one-shot games (this is done as for Table 5 in section 2).

Cooperation is stable across supergames for groups of 4 and somewhat larger than what could be expected in repeated games with fixed pairs. Data from only one supergame is available for groups of 14. Cooperation levels are smaller than for groups of 4 and closer to one-shot levels. The result from groups of 4 leads the authors to conclude that cooperation can be sustained even in anonymous settings with personal histories.⁶⁰ Note that even though the cooperation rate is high with random matching, the PD payoffs and discount parameters are such that even in a one-shot game one would expect a relatively high cooperation rate. As such, much of the cooperation with random matching may not necessarily pertain to community enforcement. In fact, Camera et al. [2012] estimate a modified version of the SFEM methodology and report that 34% of strategies (by far the most popular strategy in their data) correspond to AC. This is much higher than anything reported in

⁵⁹This result is even more surprising once an element of the experimental design in Duffy and Ochs [2009] is taken into consideration. Each session consisted of 6 or 14 subjects depending the size of the group; in other words, all the subjects in one session play together in all supergames. As such, after the end of a supergame with random matching a new supergame with random matching would start with the same set of players. Therefore, the likelihood of future interaction is greater than the intended probability of continuation. A similar caveat applies to Schwartz et al. [2000] who study cooperation under anonymous random matching in a modified prisoners' dilemma. They also find very low levels of cooperation when subjects have no information about the past behavior of their partners.

⁶⁰Bigoni et al. [2013] presents data from a similar experiment using university staff instead of students as a sample. They find cooperation rates for staff members which are smaller than for students but still greater than could be expected from one-shot games.

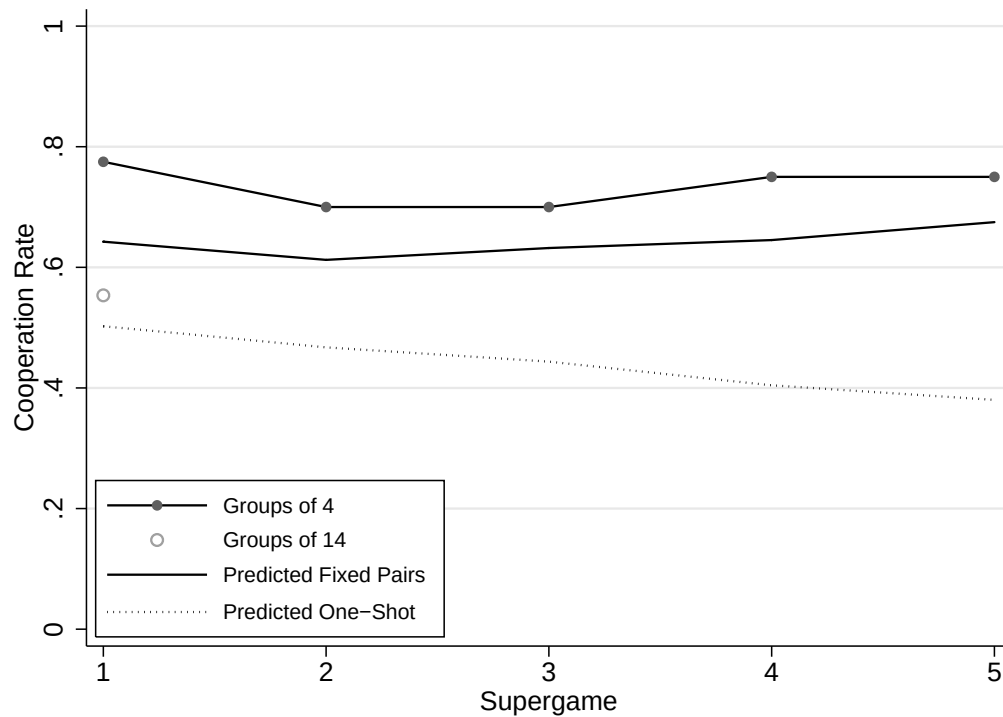


Figure 8: Round 1 Cooperation in Anonymous Random Matching Games with Personal Histories - data from Camera and Casari [2009]

Table 10 for perfect monitoring in fixed pairs and is not a strategy that implements community enforcement. They do find, however, that 34% of strategies correspond to either Grim or TFT.

In conclusion, while more research needs to be done for different combination of parameters (g, ℓ, δ) and group sizes, the current evidence suggest that cooperation will not arise under anonymous random matching with personal histories unless the group size is small.

Result 11 *Cooperation is unlikely to arise in groups with anonymous random matching unless payoffs are conducive to cooperation and the group is small.*

This result somewhat limits the applicability of contagion equilibrium as

it is unlikely that very small groups will face anonymous interactions.

Can community enforcement lead to more cooperation in less informational stringent environments? Several papers have looked at how information can help community enforcement.

Schwartz et al. [2000] study cooperation in an infinitely repeated modified prisoners' dilemma game with random matching and find that providing information about the past behavior of the partners significantly increases cooperation levels. The reason is that subjects are more likely to cooperate if their partner has cooperated before, providing incentives to cooperate.

Duffy and Ochs [2009] also study the impact of information on cooperation in random matching games. They consider two information treatments in addition to the treatment with personal histories discussed above. In one of these treatments subjects are informed of the average payoff from the previous match of the current partner. In another treatment subjects are informed of the last action chosen by the current partner. They find that cooperation is greater with information but the difference with the treatment with personal histories is only significant for the average payoff treatment. They conclude that the type of matching is a more important determinant of cooperation than the information available.

Camera and Casari [2009] also present experimental results from treatments with varying information in addition to their personal histories treatment discussed above. They have two information treatments. Both treatments involve subjects observing all the actions taken by the four members of the group while in one treatment identities were not revealed, while in the other treatment the subjects knew who did what. We call these two treatments "anonymous public histories" and "perfect information."

Figure 9 shows the evolution of round one cooperation for these three treatments. While in the first supergame there are no significant differences across the three treatments, by the last supergame cooperation is significantly greater under perfect information than under anonymous private monitoring and anonymous public monitoring. In summary, being able to identify who defected and who cooperated seems more important to support cooperation than simply knowing the distribution of actions without identities.

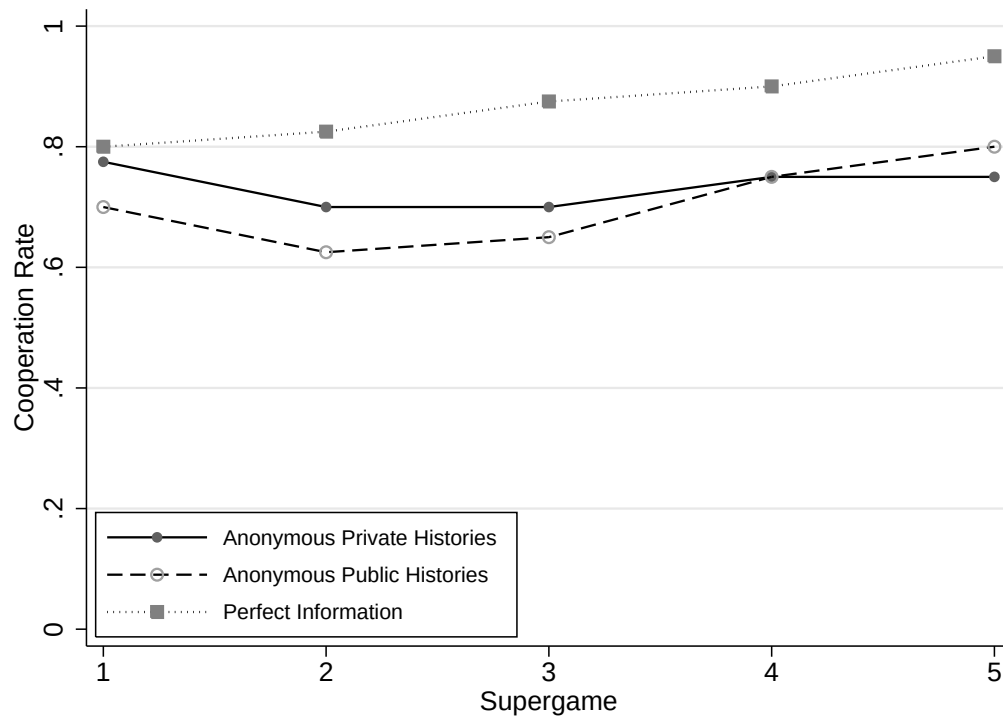


Figure 9: Round 1 Cooperation by Information Treatment - data from Camera and Casari [2009]

Duffy et al. [2013] study the impact of providing information on past behavior in random matching infinitely repeated trust games. They find that providing information significantly increases trust and trustworthiness, consistent with the previously discussed evidence from prisoners' dilemma games.

The treatment with perfect information in Camera and Casari [2009] provides an opportunity to study whether subjects rely on community enforcement or personal enforcement to support cooperation. In that treatment subjects could support cooperation in equilibrium by relying on personal enforcement alone: since there is no anonymity, subjects could consider the interactions with each other subject as a different repeated game and use some retaliatory strategy like Grim in those particular interactions. Subjects could still rely on community enforcement of some type. For example, subjects could

Table 12: Community vs. Personal Enforcement under Perfect Information in Camera and Casari [2009]- Round 2 Cooperation

	Action in round 1 of partner in round 2	
	D	C
Different Round 1 and 2 Partner	61.90	88.79
Same Round 1 and 2 Partner	40.00	95.52

stop cooperating if any member of the group had defected before (all group members are punished). Alternatively, subjects could stop cooperating with subjects who had deviated before (only deviators are punished).

To help us understand the type of enforcement used by subjects, Table 6 shows the percentage of cooperation in round 2 as a function of whether the partner in round 2 is the same as in round 1 and as a function of the action in round 1 of the partner in round 2. We focus on round 2 behavior as later behavior of others is likely to be affected by own behavior, making comparison difficult to interpret.

When the partner in round 2 is different than the partner in round 1, subjects are more likely to cooperate in round 2 with a partner who has cooperated in round 1 (88.79 percent against 61.90 percent). This suggests that subjects do not consider the relationships in isolation, but use information on what others have done to others when deciding what to do.⁶¹ Note that, the difference in behavior is greater when the partner in round 1 and 2 is the same subject (95.52 percent against 40 percent). Hence, subjects do not rely solely

⁶¹Kusakawa et al. [2012] also provides evidence consistent with community enforcement. They consider an environment in which players one and two play a one-shot prisoners' dilemma and then one of the two players is randomly chosen to play an infinitely repeated PD game with player three. They find that player three is significantly more likely to cooperate with a player who cooperated in the one-shot game. Expecting this, players one and two are more likely to cooperate in the one-shot game when player three observes the behavior from the one shot game relative to a treatment in which player three does not observe that behavior. This line of research should be continue so that we can better understand the power of community enforcement outside random matching games.

on community enforcement but use personal enforcement as well. Subjects respond more to a defection that they themselves have suffered than one suffered by another player. This can help explain the greater levels of cooperation observed under perfect information relative to public anonymous monitoring.

Result 12 *In small groups, community enforcement can be used by subjects to support cooperation and this is facilitated by providing information about past behavior. In particular, information that allows to punish deviators facilitates cooperation.*

More research is needed to understand the robustness of community enforcement in larger groups and the working of alternative institutions that may provide information. For example, to which degree would it help to have mechanisms relying on voluntary reporting by players of their past experience? Who would report and how would subjects use these reports?

7 Other stage games and dynamic games

The literature has studied how the “shadow of the future” can limit opportunistic behavior in other environments besides repeated PDs. We have already mentioned in passing some of the other games that have been studied. Closest to the PD, there are papers with a PD structure but with more than two choices. These include Holt [1985], where subjects play a Cournot duopoly game where they can select one of 18 output quantity, as well as Schwartz et al. [2000] (with random-rematching), Dreber et al. [2008], and Embrey et al. [2013] (the last two both with imperfect public monitoring) that use games with a PD structure that offer a third choice. Palfrey and Rosenthal [1994] study the voluntary contribution mechanism. Although these games are all different from the basic PD, many of the results overlap. Some distinct results these papers identify are that: 1) The availability of a costly punishment option increases overall cooperation rates. However, on average payoffs are not higher in such a situation. Hence, when punishments are available, those who do not use it and cooperate do best. 2) The availability of a “somewhat” cooperative option (which is predicted to be selected by renegotiation proofness

in this context with imperfect public monitoring) is not popular, i.e. subjects in that experiment prefer to either defect or cooperate at the highest possible level.

There are other papers looking at different stage games. Engle-Warnick and Slonim [2004], Engle-Warnick and Slonim [2006a], and Engle-Warnick and Slonim [2006b] study repeated trust games as discussed in section 2. Bernard et al. [2013] study an infinitely repeated gift-exchange game. Their parameters are selected such that the returns to sending a gift are modest (as opposed to the typical gift-exchange experiments). With such parameters, it has been shown that gift-exchange is much lower in a typical experiment without random termination. In their setting, gift-exchange can be an equilibrium phenomenon, and indeed they observe gift-exchange, despite the lower than usual potential returns to gift giving. Cason and Mui [2014] study a three players game called the Divide-and-conquer Coordinated Resistance Game. In that game a *Leader* can attempt a transgression against one or both of the *Responders* who can either acquiesce or challenge. A successful transgression against a responder takes money away from him. Part of that money is destroyed, the rest goes to the leader if he transgresses against both responders, or is shared (with more going to the leader) if the transgression is only towards one responder. Challenging (by the responders) is costly and it is only successful if both responders do it. The experiment explores the role of repetition and communication on rates of transgression. This setting is quite different from experiments such as the PD because the one-shot game has multiple equilibria (and so does the finitely repeated game). Their findings indicate that repetitions decrease the rate of transgressions, but that this is equally true for finitely repeated and randomly terminated games. Finally, Cabral et al. [2014] study a veto game as discussed in section 4.

While there is still much to be understood using the PD, clearly there is enormous scope for work exploring different stage games with random termination to better understand how repetition may affect behavior and connect this literature with particular applications. For example, the aforementioned paper by Bernard et al. [2013] studies repeated gift-exchange to answer questions related to the labor market.

There is another direction of study that does not differ in the stage game but in the fact that the stage game changes over time. Infinitely repeated games are only a small subset of dynamic games and new research has started to explore situations where choices can be conditioned on a state. These fall in two categories, those where the transitions between stage games is determined exogenously and those where they are a function of the agents' choices.

Rojas [2012] experimentally investigates the ability of subjects to support collusion in an environment à la Rotemberg and Saloner [1986]. Every round demand can be low, medium, or high, and the demand state is known to subjects before they choose their quantity (low or high). The design is such that both main parameter specifications support an equilibrium where the temptation to deviate is too important and equilibrium behavior dictates that agents must produce the high quantity when demand is high but can collude for other demand levels. The data indicates that subjects can indeed cooperate (collude) in such an environment. In addition, as predicted, subjects are less likely to cooperate (collude) when subjects learn that demand will be high. When demand comes back to a lower level, subjects are more likely to cooperate. This clearly indicate that subjects react in sensible ways to dynamic incentives even in more complex settings. Exogenous transitions across states does not destroy the subject's ability to cooperate.

Kloosterman [2013b] also investigates situations where the transitions between games is exogenous, but his focus is on the implications of beliefs and signals about the future. In a first set of experiments that uses two stage-game PDs, subjects first play one of the two games, and then, with a certain probability they either play the same PD or a different one in all subsequent periods. The design varies which PD a subject first plays and the probability that the PD changes or stays the same for the rest of the supergame. The parameters are such that cooperation in the first choice should depend on how likely the PD which is more favorable to cooperation is in the future. Indeed the data supports this idea.

The other set of experiments tests how subjects react to imperfect signals about the future conditions. In this case the stage game is an asymmetric partnership game where the *project* is successful if and only if both subjects

exert effort. The asymmetry arises because only one of the two subjects will receive the benefit of the project. Who is the beneficiary in the current period is known, but the signal gives (imperfect) information about the beneficiary in the next round. The experiment varies the precision of the signal and the payoffs to test the surprising prediction from Kloosterman [2013a] that increasing the precision of information, in this case, makes cooperation more difficult to support. Much of the data is in line with implications of the theory (reaction to signals, reaction to payoffs, etc.). However, cooperation is higher, not lower, when the signal is more precise. The explanation given for this discrepancy is based on the size of the basin of attractions, and thus suggests that the earlier results on the role of basin of attractions in the infinitely repeated PD may extend beyond the simple deterministic and stationary environment.

A recent sequence of papers explores the extent to which behavior in relatively complex dynamic environments where transitions depend on past choices is consistent with the prediction of Markov perfect equilibria. Battaglini et al. [2012] explores a repeated legislative bargaining game where in every round an endowment can either be invested in a durable public good or consumed (pork). The experiment studies the impact of the voting rule: unanimity, majority, or dictatorship. In theory, a higher voting requirement generates more efficiency. Indeed, the data supports this view and there is more public good as more votes are required for support. Battaglini and Palfrey [2012] also explore a repeated legislative bargaining game without public good. The state variable in this case is the status quo (which takes effect if the proposal on the floor is rejected) which is the last approved proposal. They conduct multiple treatment manipulations: concavity of the utility function, the availability of a condorcet alternative, and the discount factor. The treatment effects are in the direction suggested by the Markov perfect equilibrium. Finally, Battaglini et al. [2013] study a repeated voluntary contribution mechanism where the public good is durable. They vary whether investments are reversible or not and the number of subjects in the group. The Markov perfect equilibrium has higher levels of public good when investments are irreversible. The data is consistent with that prediction. On the other hand, there is a tendency for over-accumulating the public good early on, a trend that can be corrected later on in the reversible

case, but not in the irreversible case.

These last three papers show many comparative statics consistent with the predictions of Markov perfect equilibrium. However, in a simple infinitely repeated PD, the unique Markov perfect equilibrium involves defection in every round, which we have shown is not what happens in such a game. Hence, a natural question seems to be, when is the Markov assumption a reasonably good predictor of behavior? Or even, is there evidence that subjects play Markov perfect strategies? To investigate such questions, Vespa [2014] studies a simple version of the dynamic commons problem. Each round, subjects can extract resources from a common pool which replenishes at a given rate. Cooperation involves extracting less than is optimal in a one-shot game to let the resource grow. He designs an environment that allows him to identify not only the Markov perfect equilibrium payoff but also the fully efficient symmetric equilibrium payoff.⁶² Much of the data is consistent with Markov behavior. However, when opportunities for opportunistic behavior are constrained, which in this setting is also equivalent to a simpler stage game, a substantial fraction of subjects aim to support full cooperation. The percentage of strategies supporting full cooperation in those treatments is comparable to those observed in PD experiments with similar payoff parameters.

In many situations, agents who do not find a way to cooperate in a relation will prefer to exit the relation altogether instead of defecting for ever. Wilson and Wu [2014] study how such a possibility affects choices and what the impact of the outside value is on behavior. This offers a new and interesting window on the potential role of contracts and the legal system as those determine what happens when relations are severed outside of the laboratory. The experiment involves many treatments all based on the partnership game, some where the outside value is fixed and some where the outside value depends on the behavior of the agent while in the relation. As predicted by theory, they

⁶²The focus on Markov perfect equilibria in dynamic games is often a matter of convenience as the environments are too complex to solve for other equilibria. Hence, one must be careful making the jump from observing comparative statics in line with the Markov prediction to concluding that subjects use Markov strategies. Designing a setting where we can solve for other equilibria gives more force to the conclusions, either positive or negative.

observe much more termination once the outside option exceeds the in-relation minmax. The possibility of termination seems to increase the use of strategies that start by cooperating. Also, punishment is more likely to take the form of termination (rather than simple defection). However, subjects become more lenient: they need more bad signals before triggering a punishment. One rule stands out among the asymmetric termination rule: efficiency is highest when the subject that cooperated the most in the relation is treated more favorably in case of termination.

These papers all point in different but extremely interesting directions for future research. Moving toward dynamic games opens the door to many applications of interest. However, the frequent requirement to focus on Markov perfect equilibria means that a better understanding of when such an assumption is reasonable is important. Finding concepts and regularities that connect these more complicated environments back to the simpler case of the PD will probably yield important rewards moving forward.

There are other types of dynamic games that, although distinct in the way they are analyzed, seem to be naturally connected and we think that future research exploring the similarities, connections, as well as the differences will be useful. In particular, the recent studies of games in continuous time would seem related. Although the theoretical tools to analyze them are different, they too generate cooperation in social dilemmas where no such cooperation is predicted in the one-shot game. Two recent experiments on games in continuous time are Friedman and Oprea [2012] and Bigoni et al. [2013]. Both show that subjects can support cooperation under continuous time. However, surprisingly, in continuous time, Bigoni et al. [2013] show that cooperation rates can be lower with a randomly determined horizon as opposed to a deterministic horizon. Understanding the behavioral sources of any differences or similarities could prove enlightening for theoretical work. This might come through better understanding of how strategy choices are affected by the details of the strategic environment.

8 Conclusions

The theory of infinitely repeated games has shown that repetition can help people overcome opportunistic behavior and lead to cooperation. While this idea has been applied in many fields of economics and other social sciences, only recently a large experimental literature has considered its validity. In addition to surveying the existing experimental literature, we gather and analyze available experimental data in order to focus on robust results.

We observe that *the shadow of the future* can lead to cooperation and that theory helps understand under what conditions this will occur. Moreover, the experimental evidence provides a solution to the multiplicity of predictions provided by theory. We find that people do not necessarily coordinate on the best equilibrium. Cooperation is low when it is not robust to strategic uncertainty and increases with robustness to strategic uncertainty. We observe high levels of cooperation only when the parameters are such that attempting to cooperate is not too risky. However, as discussed in section 2, more work is needed to find the index that best predicts cooperation.

We also see that subjects tend to use simple strategies to punish defection under perfect monitoring. While the fact that subjects do use punishments to support cooperation is consistent with theory, one of the most used strategies (TFT) involves non-credible punishments. As such, this raises the questions of why subjects use it instead of other strategies with better theoretical properties. More work is also needed to understand how the use of strategies changes with the parameters of the game.

We also see that *the shadow of the future* can lead to cooperation under imperfect monitoring (both public and private). Much more work is needed to better understand cooperation under imperfect monitoring (especially under private monitoring) and its robustness to the specifics of the environment. In particular, the literature should pay attention to how details of the monitoring technology affects the capacity to coordinate on cooperation and the types of strategies used. For imperfect public monitoring, we see that subjects are more likely to use lenient and forgiving strategies relative to perfect monitoring. More work is needed on the strategies that subjects use under imperfect private

monitoring and the connection to those used in theoretical work.

We also see that subjects are able to support cooperation when partners change from period to period. However, so far this seems true only in very small groups. It seems fair to say that it is much more difficult for subjects to support cooperation when they are randomly rematched than when they play in fixed pairs. In fact, it seems easier for subjects to cooperate even if monitoring is imperfect, but they are in fixed pairs, than if they are randomly rematched. Still more work is needed to understand the strategies subjects use under perfect monitoring with random rematching (to better understand the relative importance of personal and community enforcement) and how other information institutions (like reputational mechanisms) may help subjects support cooperation.

Communication may be another important determinant of cooperation. As discussed by Cooper and Kühn [2014a], Andersson and Wengström [2012], Embrey et al. [2013] and Cooper and Kühn [2014b], communication could in principle have different effects on cooperation.⁶³ On the one hand, communication before the supergame could help subjects coordinate on cooperation; increasing cooperation rates. On the other hand, communication after a defection could help renegotiation and the expectation of this could result in lower cooperation rates. Cooper and Kühn [2014a] study the effects of several types of communication on cooperation in a game in which subjects play a social dilemma first and then a coordination game.⁶⁴ Among other very interesting results, they find that free-form communication (chats) leads to persistent increases of cooperation, while other forms of communication do not. Although the simplified dynamic game facilitates the study of communication, given the caveats mentioned in section 3 about this way of inducing infinitely repeated games (finite repetition followed by a coordination game), it seems appropriate to explore the generalizability of the results from Cooper and Kühn [2014a] using other ways to induce infinitely repeated games in the laboratory. More

⁶³See also Cason and Mui [2014] which studies the effect of communication in repeated divide-and-conquer coordinated resistance game.

⁶⁴The coordination game can be derived from the continuation payoffs that would arise from an infinitely repeated game in which deviations would be punished for ever by reversion to the one-shot Nash equilibrium.

generally, it is important to study the effects of communication for different combinations of parameters so as to be able to replicate the analysis of the determinants of cooperation under perfect monitoring that we have offered in section 2. Will we find high levels of cooperation even when cooperation is not risk dominant? Could it be that once we allow for communication, subjects do coordinate on the Pareto efficient equilibrium? Will subjects use different types of punishments?

While cooperation in repeated games may be affected by elements of the game like payoff parameters, probability of continuation, available information, matching or communication, it may also be affected by elements from outside of the game. For example, the literature has looked at the role of personal characteristics in explaining differences in behavior but found no robust relationship. We believe we should keep looking for personal characteristics which may help explain behavior in repeated games. Another element from outside the game that may affect behavior is history. As we have seen, the experience in a given treatment affects behavior. Similarly, experience in other treatments or other games may affect behavior as well.⁶⁵ We should study whether history is used as an equilibrium selection device in infinitely repeated games. The experiences that subjects had before entering the supergame and the ways in which subjects arrived to a supergame may affect cooperation. Similarly, local norms of cooperation may affect the expectations of behavior that subjects bring to the laboratory. This suggests that a comparative study of behavior across cultures could be of interest.

Given that the formation and termination of groups is usually endogenous, following Bernard et al. [2013] and Wilson and Wu [2014], more work is needed to see how these realistic features affect cooperation and its determinants.

While in this article we have focused on experiments that use the PD as the stage game, we have also mentioned many articles using other stage games (other social dilemmas, trust games, veto games, etc.). To test the robustness of the results, additional experiments with other stage games would be welcome. For example, it would be interesting to know how repetition affects contributions in linear public good games with a continuum or high

⁶⁵Footnote 29 in Dal Bó [2005] presents evidence consistent with this idea.

number of contributions levels. This could allow for different strategies that may reduce the risk from strategic uncertainty and support cooperation (like starting small).

Similarly, future articles on dynamic games may shed light on the types of punishments used when actions affect the state. When do subjects coordinate on Markov strategies? When do they rely on punishments that also depend on elements of the history that are not captured by the state? Can the elicitation of strategies help us answer these questions?

Much progress has been made in understanding the determinants of cooperation in infinitely repeated games. Many questions remain to be answered and we hope this article provides a clear picture of the state of the literature and hints at productive lines of future research.

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